

REVIEW

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Review of radiolarian microfossils as a tool for reconstructing sea surface temperature of the past in the Northwest Pacific

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Abstract

In this review we re-evaluated the potential of radiolarian species as palaeoceanographic proxies in the Northwest Pacific Ocean relying on 33 new samples collected since 2021, combined with already published datasets. Our re-evaluation revealed significant differences between Sea of Japan and Northwest Pacific radiolarian assemblages, leading to the exclusion of the Sea of Japan dataset for further sea surface temperature (SST) reconstructions in the Northwest Pacific. By employing factor analyses, we were able to identify four distinct radiolarian assemblages characterizing the Northwest Pacific and East China Sea as follows: the Subtropical, the Sea of Okhotsk-related sub-arctic, the Oyashio Current to transitional zone-related, and the coastal water assemblages. Each assemblage showed specific species associations with different water masses and SST ranges. Species like *Tetrapyle circularis/fruticosa* and *Dictyocoryne tetrathalamus* were associated with waters above 24 °C, while *Lithomelissa setosa* and *Ceratospyris borealis* were associated temperatures below 14 °C. Based on a review of the literature about modern radiolarian species blooming conditions in the Northwest Pacific, we suggest that radiolarian species-based reconstructed SSTs more likely infer summer SSTs. Applying weighted averaging partial least squares transfer function on selected radiolarian species showing strong affinities with changes in summer SST, we re-evaluated past summer SSTs at IODP Site U1429 in the northern East China Sea with high accuracy ($R^2 = 0.97$, error margin ± 1.4 °C). Our radiolarian-based summer SST reconstruction is similar to *Globigerinoides ruber* Mg/Ca-based summer SSTs, though showing minor disparities during glacial periods, while constant disparities were observed with alkenone-based SST estimates at the same site, which is likely due to seasonal biases. Notably, we identified *L. setosa* as a potential proxy for East Asian Winter Monsoon intensity.

Keywords Radiolarian, Sea surface temperature (SST), Radiolarian-based SSTs, *G. ruber* Mg/Ca-based SSTs, Alkenone-based SSTs, Northwest Pacific, Water masses

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1 Introduction

The Earth's climate is currently undergoing significant warming, with temperatures increasing by 1.25 °C between 1960 and 2020 and projections suggesting a potential 2 °C rise by 2100 (IPCC 2023). This warming affects the complex tropical monsoon system that controls East Asian climate and Northwest Pacific oceanography. These changes have major implications for a large part of the world's population and thus understanding how Earth's climate system responds to these new conditions is crucial. Reconstructing environmental conditions can offer valuable insights into our future climate.

Various proxies, including planktic foraminifera, radiolarian fossils, and calcareous nannofossils, are used to reconstruct past environmental conditions. One commonly used proxy for reconstructing past climates is paleo-sea surface temperature (SST), as it is highly sensitive to changes in the Earth's orbital configuration that regulate heat balances. Many paleothermometrical proxies rely on geochemical analysis of planktic foraminifera shells (e.g., Clemens et al. 2018; Kubota et al. 2010; 2019; Sagawa et al. 2023). However, the preservation of foraminifera fossils can be compromised when deposited at depths near the carbonate compensation depth (CCD), where calcareous matter is dissolved. The CCD is shallower in the North Pacific Ocean compared to other ocean basins, such as the North Atlantic, making it challenging to reconstruct past SSTs using foraminifera in this region (e.g., Berger et al. 1982). As a result, siliceous microfossil assemblages may provide an alternative for estimating SSTs in the North Pacific Ocean.

Among siliceous plankton groups, radiolarians have emerged as valuable proxy for reconstructing sea surface temperatures (SSTs) in the North Pacific region since the 1990 s. While numerous studies have demonstrated their suitability for paleoceanographic reconstructions (e.g., Pisias et al. 1997; Kamikuri et al. 2008; Matsuzaki et al. 2014; Matsuzaki and Itaki 2017; Hernandez-Almeida et al., 2017), significant questions remain regarding the seasonal and depth-specific nature of radiolarian-based temperature signals, as well as their reliability compared to well constrained geochemical SST proxies. In such a context, after having reviewed the current state of major radiolarian research dealing with modern species spatial distribution and past attempts in SST estimates, we propose a comprehensive review of radiolarian-based SST reconstructions in the Northwest Pacific, focusing on three key aspects: (1) the spatial distribution patterns of key species used in SST estimations, (2) the methodological strengths and limitations of radiolarian-based reconstructions, and (3) a critical comparison with geochemical SST proxies.

1.1 Polycystine radiolarians and their spatial distribution

Radiolarians are planktonic marine Protista that are found throughout the world's oceans. They also constitute a sister group to foraminifers within Rhizaria (Suzuki and Not 2015; Lazarus et al. 2021). A key characteristic of radiolarians is that they possess mineral skeletons composed of SiO₂ or SrSO₄; however, only skeletons containing SiO₂ (Polycystina) are preserved in the geological record because SrSO₄ skeletons (Acantharia) dissolve during diagenesis at the sediment–water interface (e.g., Anderson, 1983; Suzuki and Not 2015; Lazarus et al. 2021). Thus, radiolarians in the class Polycystina, which have amorphous opaline silica skeletons (SiO₂·nH₂O), are the most studied group as their skeletons are abundant and well-preserved in deep-sea sediments (e.g., Kling 1979; Anderson, 1983; Lazarus et al. 2021).

Polycystine radiolarians are distributed throughout the water column, from shallow to deep water, and have been extensively studied in various regions of the North Pacific. These regions include the Kuroshio area of the Northwest Pacific (e.g., Ishitani et al. 2008; Itaki et al. 2010; Matsuzaki et al. 2016, 2020a), the northern Sea of Japan (Itaki 2003), the Sea of Okhotsk (Nimmergut and Abelman 2002; Okazaki et al. 2004), the Oyashio area of the Northwest Pacific (Okazaki et al. 2004), the central North Pacific (Renz 1976; Kling 1979), and the western and central equatorial Pacific (Yamashita et al. 2002). The vertical distribution of radiolarian species in the North Pacific is thus well understood. To summarize the most important takeaways from previous cited studies, in the shallow warm, saline waters of the North Pacific (0–200 m), species such as *Didymocystis tetrathalamus* (Haeckel) and the *Tetrapyle circularis/fruticosa* group, as defined by Zhang and Suzuki (2017), are abundant. At subsurface depths (200–500 m), abundant species include *Cycladophora bicornis* (Popofsky) and *Spongodiscus cf. resurgens* Ehrenberg, while at intermediate depths, subarctic species like *Cycladophora davisiana* Ehrenberg and *Tholospira weddellium* Lazarus, Faust, and Popova-Goll dominate (Matsuzaki et al. 2016, 2020a). Exclusive to deep waters of the North Pacific (> 1000 m), species like the *Tricolocapsa papillosa/diadema* group and the *Cornutella profunda* Ehrenberg group dominate the radiolarian assemblage (Matsuzaki et al. 2020a).

Numerous studies have also clarified the geographical distributions of modern radiolarian species, promoting their use as a palaeoceanographic index (Lombardi and Boden 1985; Yamauchi 1986; Motoyama and Nishimura 2005; Kamikuri et al. 2008; Boltovskoy et al. 2010). Matsuzaki and Itaki (2017) introduced a new dataset for estimating past SSTs and intermediate water temperatures in the Northwest Pacific. Matsuzaki and Itaki (2017) also reviewed the geographical distributions of 47 radiolarian

species, which have relative abundances higher than 2% at least at one location in the Northwest Pacific. For instance, species such as *T. weddellium* are associated with colder intermediate-to-subsurface waters with a temperature of around 7 °C in the North Pacific, whereas species such as *Lithelius minor* Jørgensen are associated with relatively warmer intermediate waters. In addition, species such as *C. davisiana* were found in the intermediate water of the Sea of Okhotsk and neighboring regions. In regions such as the semi-isolated Sea of Japan, the vertical and spatial distributions of radiolarian species are distinct from those of the North Pacific due to the limited exchange of intermediate water between the Sea of Japan and the North Pacific. Previous research by Itaki (2003) and Motoyama et al. (2016) provided insights into the vertical and geographical distributions of radiolarian species through analyses of sediments cores, surface sediments, and plankton tows. Notably, species such as *Tetrapyle circularis/fruticosa* were abundant near the Tsushima Strait, influenced by the Tsushima Warm Current, a branch of the Kuroshio Current. Conversely, the subarctic species *Pseudodictyophimus gracilipes* (Bailey) was abundant in the northern region of the Sea of Japan. However, the main differences between the North Pacific and the Sea of Japan are observed in the intermediate and deep-water masses. For example, *Actinomma leptodermum* (Jørgensen), which is common at intermediate and deep-water depths in the Sea of Japan, is rarely found in the Northwest Pacific. This distribution pattern suggests that *A. leptodermum* could potentially serve as an indicator species for the intermediate and deep-water masses of the Sea of Japan, known as Japan Sea Proper Water (JSPW) (Itaki 2003; Motoyama et al. 2016).

1.2 Radiolarian-based sea surface temperature estimates in the Northwestern Pacific

The sensitivity of radiolarians to oceanographic changes has led to their use in reconstructing past sea surface temperatures (SSTs). The method using siliceous microfossils, including diatoms and radiolarians, for this purpose has been developed over several decades. Among them, the Climate: Long-range Investigation, Mapping, and Prediction (CLIMAP) project reconstructed worldwide SSTs since the Last Glacial Maximum, incorporating these microfossils in regions with sediments showing low carbonate and high silicate preservation, such as the Southern and Pacific Oceans (CLIMAP, 1976; Moore 1978; Moore et al. 1980). The first radiolarian-based SST estimates were conducted by Nigrini (1970), who introduced a radiolarian temperature number known as the “Tr” ratio. Sachs (1973) compiled a dataset of radiolarian assemblages from surface sediments and correlated it with modern summer SSTs. Further, a transfer function

for estimating paleo-SSTs was developed using Q-mode factor analysis, based on the statistical approach of Imbrie and Kipp (1971).

Sachs (1973) was the first to propose a radiolarian dataset covering the entire North Pacific for estimating past summer SSTs using an Imbrie and Kipp (1971) transfer function; these estimates were later updated by Pisias et al. (1997) using a similar method. The transfer function developed in the latter study was used to estimate past annual SSTs with a standard error of 1.63 °C. For the Central North Pacific, Kamikuri et al. (2008), relying on the preliminary works of Motoyama and Nishimura (2005), also introduced a transfer function for estimating past winter and summer SSTs and achieved a standard error of less than 1 °C. Matsuzaki et al. (2014) compiled a dataset for the northwestern Pacific by extracting shallow-water radiolarian taxa abundances from the datasets of Pisias et al. (1997), Itaki (2003), and Kamikuri et al. (2008); this gave reconstructed annual SSTs with an error of less than 1 °C. However, a major concern about these studies is that the standard errors of the reconstructed annual SSTs were not cross-validated, which led to small error margins but is subject to overfitting.

In recent years, two additional radiolarian datasets for estimating past SSTs in the northwestern Pacific have been published. The first of these was developed for the tropical Northwest Pacific with broad coverage in the South China Sea, comprising 160 samples (Hernandez-Almeida et al., 2017). Hernandez-Almeida et al. (2017) suggested that, in the tropics, radiolarians are more sensitive to winter SSTs due to higher primary productivity. By applying locally weighted averaging regression with inverse deshrinking, Hernandez-Almeida et al. (2017) were able to reconstruct past SSTs with a Root Mean Square Error of Prediction (RMSEP) of 1.6 °C. Later, Hernandez-Almeida et al. (2020) harmonized a published modern radiolarian dataset for the entire Pacific Ocean and developed a new calibration method, which was then applied to published radiolarian census counts and obtained from 31 cores in the Bering Sea, West Pacific marginal seas, and the southwestern and eastern equatorial Pacific, covering the last 165 ka. Based on comparisons with previously published radiolarian-based and geochemical-based SST reconstructions, Hernandez-Almeida et al. (2020) also suggested that radiolarian-based reconstructions provide more accurate temperature estimates than geochemical methods based on, for example, alkenone during glacial periods in the Sea of Japan, Bering Sea, and Sea of Okhotsk.

Another radiolarian-based SST dataset for the area near the coast of Japan was developed by analyzing 88 surface sediment samples retrieved by the Geological Survey of Japan and the Japanese Agency for

Marine-Earth Science and Technology (JAMSTEC) since the 1970 s (Matsuzaki and Itaki, 2017). Matsuzaki and Itaki (2017) also made SST estimates for the East China Sea by selecting only species inhabiting the upper 200 m of the water column based on plankton tow observations (Matsuzaki et al. 2016). In addition, because at middle latitudes the flux of warm species such as the *Tetrapyle circularis/fruticosa* group is highest in summer and autumn (Itaki et al. 2008), Matsuzaki and Itaki (2017) suggested that radiolarian-based SSTs tend to be related with summer conditions and were able to reconstruct past summer SSTs with an error of 0.75 °C. However, unlike Hernandez-Almeida et al. (2017), standard errors were not cross-validated again, and thus it is highly probable that the reconstructions are overfitted. Using the same modern reference dataset described by Matsuzaki and Itaki (2017), Matsuzaki et al. (2020b) proposed a provocative but simple paleotemperature equation for reconstructing SSTs over the last 10 million years by selecting only modern species that were extant since the Late Miocene. While standard errors were not cross-validated, Matsuzaki et al. (2020b) used an adjusted R^2 approach to provide more accurate error estimates. This method addresses key limitations of the standard R^2 , in particular its susceptibility to overfitting. A very recent application of this method (Matsuzaki 2023) to the central Northwest Pacific at Ocean Drilling Program (ODP) site 1208 showed that the equation of Matsuzaki et al. (2020b) could indeed be used to reconstruct SSTs over the last 10 million years and that the reconstructed SSTs were sensitive to all known global climate events.

1.3 Issues still to be addressed and objectives of the study

Since the mid-2010 s, radiolarian-based palaeoceanographic studies have seen significant advancements. However, concerns remain, particularly regarding the open Northwest Pacific. Despite the pioneering efforts of Moore (1978), Moore et al. (1980), and Pisias et al. (1997), there has been a lack of recent data for this area. Such limitation in data has resulted in a coastal bias in new several datasets. For instance, in Matsuzaki and Itaki (2017) dataset, samples were predominantly collected within Japan Exclusive Economic Zone (EEZ) by the Geological Survey of Japan and Japan Agency for Marine Science and Technology (JAMSTEC). The Japan EEZ, influenced by the Oyashio and Kuroshio Currents, including currents like the Tsushima Warm Current, provides an optimal research environment for understanding radiolarian species distributions in relation to these currents and their correlation with sea surface temperature (SST) changes. However, the scarcity of offshore data points can restrict the broader applicability of this dataset.

The Northwest Pacific Ocean encompasses various marginal seas such as the East China Sea, the Sea of Japan, and the South China Sea. While research on radiolarians in the South China Sea has been well-documented and validated by studies revealing a primary control of radiolarian species by winter monsoon upwelling rather than SST variations alone (Hernandez-Almeida et al., 2017; Qu et al. 2020; Zhang et al. 2020), investigations into SST influences on radiolarian species are notably lacking for the East China Sea and the Sea of Japan. In addition, comparisons between radiolarian-based SST estimates and geochemical proxies have also been limited while it can strengthen the suitability of radiolarian-based SST reconstructions. Only four studies to date have addressed such concerns:

- (1) Pisias et al. (1997) and Hernández-Almeida et al. (2020) compared radiolarian-based SSTs with alkenone-derived SSTs for the Late Pleistocene,
- (2) Matsuzaki et al. (2023) evaluated radiolarian-based SSTs against TEX86-derived SSTs for the Middle Pleistocene,
- (3) Matsuzaki (2023) compared radiolarian-based SSTs with alkenone-derived for the past 10 million years.

These results showed that for high latitudes of the North Pacific, radiolarian-based SSTs are even more suitable than alkenone SSTs (Hernandez-Almeida et al., 2020). However, we still need a better understanding about each proxy limitations for example their potential biases related to the inferred water depths and seasonal variations affecting the blooming of organisms used in SST estimates.

Recent oceanographic expeditions have significantly expanded the available dataset for radiolarian studies in the Northwest Pacific. As part of the GEOTRACES program, designed to enhance understanding of ocean biogeochemical cycles, a major latitudinal transect was conducted along 150°E from 1°S to 50°N in 2022 (Obata et al., 2022). This survey (Japan Joint Usage/Research Center for Atmosphere and Ocean Science Expedition KH-22-7) resulted in the retrieval of 16 new core-top sediment samples. In addition, several expeditions took place in the East China Sea (expeditions KH-21-3, KH-22-3, and KS-22-4), during which 14 samples were collected, as well as off the Boso Peninsula and Hokkaido Island (expedition KS-24-6) and around the Japan Trench (expedition YK-21-07). These combined expeditions since 2020 have collected 33 new core-top sediment samples, significantly expanding the available dataset. This new material provides an opportunity to enhance our understanding of offshore Northwest Pacific radiolarian distributions and their relationship with sea surface

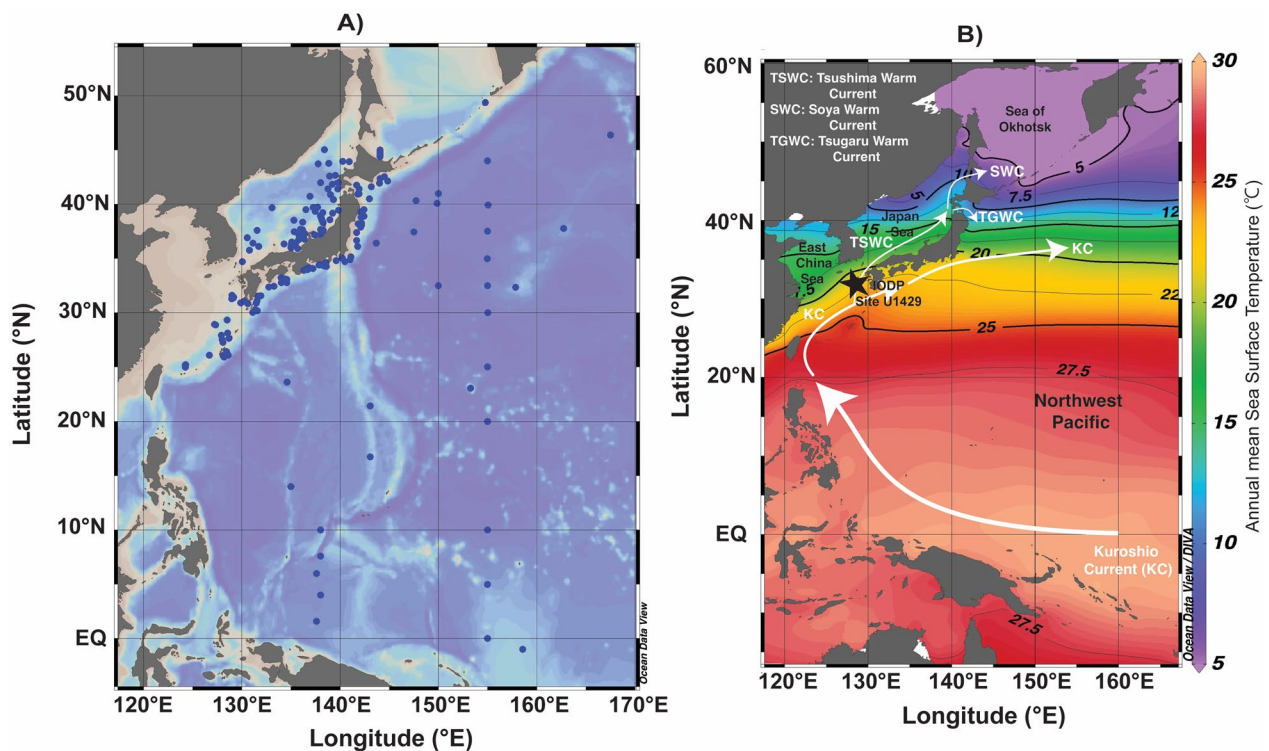


Fig. 1 In the **A** panel, the blue dots mark where sediment core tops were collected for the compilation. These core tops were utilized to create a Northwest Pacific core-top dataset. In the **B** panel, a map depicting the major currents in the researched area is shown. The star indicates the position of IODP Site U1429, while the annual sea surface temperature distribution in the Northwest Pacific, as detailed in the World Atlas Ocean by Locarnini et al. in 2018, is represented on the Z axis. The maps are sourced from ODV4 by Schlitzer et al. (2022)

temperature (SST) variations. Furthermore, the integration of East China Sea samples with Matsuzaki and Itaki (2017) dataset will enable a more comprehensive analysis of radiolarian distributions in this region. Additionally, Motoyama et al. (2016) previously uncompiled study of 57 surface sediment samples from the Sea of Japan offers valuable data.

Therefore, this review integrates new radiolarian datasets from 33 samples collected since 2020 with those already published from the Sea of Japan, Northwest Pacific, and East China Sea to achieve three primary objectives: (1) a comprehensive understanding of radiolarian species distributions across the Northwest Pacific region; (2) evaluate the relationship between key species and changes in SST; and (3) assess both the potential and limitations of this expanded modern reference dataset for SST reconstructions in the Northwest Pacific. Through this synthesis, we aim to establish a more robust framework for radiolarian-based palaeoceanographic studies in the region.

2 Review

2.1 Brief review of Northwest Pacific oceanography

The marine environments of the Northwest Pacific exhibit distinct features that are influenced by the interaction between the subtropical and subarctic gyres (Fig. 1) (Tomczak and Godfrey, 1994). In this part of the Pacific, the subtropical gyre corresponds to the North Pacific Equatorial Current (NPEC) and the Kuroshio Current (KC); the NPEC carries warm oligotrophic water and bifurcates northward to feed the warm-oligotrophic KC (Tomczak and Godfrey, 1994). The KC flows northward along the coast of the Philippines and Taiwan then enters the East China Sea (ECS) before looping back into the North Pacific (Fig. 1). Temperature data from Ocean Data View and the World Ocean Atlas show that SSTs in the tropics peak between 27 °C and 30 °C during the summer, whereas in the ECS, they range from 27 °C to 25 °C (Fig. 1). As the KC flows along the Japanese coast, it merges with the subarctic Oyashio Current (OC) off northeastern Japan (Tomczak and Godfrey, 1994) (Fig. 1). Offshore, the extension of the KC, the so-called Kuroshio Current Extension (KCE), is cooled by westerly winds, which leads to a temperature drop from 24 °C to 14 °C between summer and winter; the thermocline

Table 1 List of samples and their geographical information and Summer SST of Locarnini et al. (2018). The stations are sorted by latitude. The sample's used samples were multicore samples, gravity core and push core samples in this study

Reference	Samples	Longitude (°E)	Latitude (°N)	Summer sea surface temperature (°C)
This study	KH-22-7-OP-MC03 0–1 cm	158.57	−1.00	29.31
This study	KH-22-7-OP-MC02 0–1 cm	155.00	0.00	29.36
Matsuzaki and Itaki (2017)	NH90-1-N54-NB17	137.60	1.60	29.12
Matsuzaki and Itaki (2017)	NH90-1-N56-NB18	138.00	4.00	29.17
This study	KH-22-7-OP-MC04 0–1 cm	154.98	4.98	29.39
Matsuzaki and Itaki (2017)	NH90-1-N57-NB19	137.60	6.00	29.22
Matsuzaki and Itaki (2017)	NH90-1-N60-NB21	138.00	7.60	29.25
Matsuzaki and Itaki (2017)	NH90-1-N62-NB22	138.00	10.00	29.23
This study	KH-22-7-OP-MC05 0–1 cm	155.00	10.00	29.29
Matsuzaki and Itaki (2017)	NH90-1-N68-NB25	135.00	14.00	29.24
This study	KH-22-7-OP-MC01 0–1 cm	143.05	16.75	29.44
This study	KH-22-7-OP-MC070-1 cm	155.00	20.00	29.21
Matsuzaki and Itaki (2017)	KR02-01 PC03-01 A	143.06	21.44	29.07
Matsuzaki and Itaki (2017)	KR13-02 PC03-01 W	153.26	23.04	28.74
Matsuzaki and Itaki (2017)	NH90-1-N79-NB30	134.60	23.60	28.52
This study	KH-21-3 PL04 0–2 cm	124.26	24.99	28.85
This study	KH-22-7-OP-MC08 0–1 cm	154.99	25.00	28.73
This study	KH-21-3 PL03 0–2 cm	124.26	25.13	28.85
This study	KH-21-3 PL02 0–2 cm	124.24	25.19	28.85
This study	KH-21-3 PL01 0–2 cm	124.29	25.21	28.85
Matsuzaki and Itaki (2017)	GH08-198	128.18	25.91	28.49
Matsuzaki and Itaki (2017)	GH08-214	128.07	26.05	28.48
This study	KH-22-3-PL08-0–2 cm	126.68	26.08	28.7
Matsuzaki and Itaki (2017)	GH08-197	128.60	26.08	28.47
Matsuzaki and Itaki (2017)	GH08-247	128.56	26.15	28.46
Matsuzaki and Itaki (2017)	GH08-248	128.45	26.41	28.45
Matsuzaki and Itaki (2017)	GK14-612	128.09	27.26	28.35
Matsuzaki and Itaki (2017)	GK14-642	127.99	27.53	28.29
Matsuzaki and Itaki (2017)	GK14-694	128.68	27.64	28.31
Matsuzaki and Itaki (2017)	GK14-591	127.91	27.91	28.20
Matsuzaki and Itaki (2017)	GK14-766	128.55	28.12	28.22
This study	KS-22-4 MC04 0–1 cm	127.32	28.88	28.59
This study	KS-22-4 MC07 0–1 cm	127.22	28.93	28.59
This study	KH-22-3-PL01-0–2 cm	128.50	29.33	28.68
This study	KH-22-7-OP-MC09 0–1 cm	155.02	30.00	27.33
Matsuzaki and Itaki (2017)	GH84-3-63	131.09	30.02	27.93
Matsuzaki and Itaki (2017)	GH84-3-169	131.57	30.19	27.88
Matsuzaki and Itaki (2017)	GH84-3-30	131.35	30.29	27.89
Matsuzaki and Itaki (2017)	GH84-3-169	130.62	30.85	27.90
Matsuzaki and Itaki (2017)	GH84-3-37	131.46	30.89	27.87
Matsuzaki and Itaki (2017)	GH84-3-263	129.89	30.90	27.66
Matsuzaki and Itaki (2017)	GH84-3-224	130.19	31.04	27.72
Matsuzaki and Itaki (2017)	GH84-3-34	131.54	31.11	27.86
This study	KH-21-3 MC06 0–1 cm	128.96	31.29	27.46
This study	KH-21-3 MC04 0–1 cm	129.05	31.33	27.46
This study	KH-21-3 MC03 0–1 cm	129.13	31.36	27.46
This study	KH-21-3 MC01 0–1 cm	129.08	31.38	27.46

Table 1 (continued)

Reference	Samples	Longitude (°E)	Latitude (°N)	Summer sea surface temperature (°C)
This study	KH-21-3 MC02 0–1 cm	129.17	31.43	27.46
This study	KH-21-3 MC05 0–1 cm	129.12	31.46	27.46
Matsuzaki and Itaki (2017)	GH83-2-320	131.69	31.55	27.63
Matsuzaki and Itaki (2017)	GH83-2-264	131.92	31.56	27.63
Matsuzaki and Itaki (2017)	IODP-346-U1429 A-1H1-27–29	129.01	31.62	27.06
This study	KH-22-7-OP-MC10 0–1 cm	157.85	32.35	26.36
Matsuzaki and Itaki (2017)	GH83-2-158	132.74	32.37	27.56
Matsuzaki and Itaki (2017)	GH83-2-109	133.27	32.44	27.52
This study	KH-22-7-OP-MC11 0–1 cm	155.00	32.48	26.3
This study	KH-22-7-OP-MC21 0–1 cm	149.98	32.50	26.71
Matsuzaki and Itaki (2017)	GH82-1-24	134.02	32.77	27.50
Matsuzaki and Itaki (2017)	GH82-1-1	134.57	32.77	27.49
Matsuzaki and Itaki (2017)	GH82-1-49	134.39	33.02	27.50
Matsuzaki and Itaki (2017)	GH82-1-51	134.57	33.07	27.49
Matsuzaki and Itaki (2017)	GH82-1-69	134.36	33.12	27.50
Matsuzaki and Itaki (2017)	GH82-1-106	133.98	33.23	27.50
Matsuzaki and Itaki (2017)	GH82-2-110	136.30	33.89	26.29
Matsuzaki and Itaki (2017)	GH82-2-49	136.69	33.97	26.28
Matsuzaki and Itaki (2017)	GH82-2-66	136.57	34.01	26.28
Matsuzaki and Itaki (2017)	GH97-258	137.23	34.08	25.83
Matsuzaki and Itaki (2017)	GH82-2-11	136.94	34.09	26.29
Matsuzaki and Itaki (2017)	GH82-2-32	136.73	34.09	26.29
Matsuzaki and Itaki (2017)	GH97-118	138.58	34.29	26.15
Matsuzaki and Itaki (2017)	GH97-126	138.46	34.30	26.35
Matsuzaki and Itaki (2017)	GH97-135	138.35	34.32	26.35
Matsuzaki and Itaki (2017)	GH97-163	137.95	34.39	26.33
Matsuzaki and Itaki (2017)	GH97-175	137.84	34.41	26.32
Matsuzaki and Itaki (2017)	GH97-80	138.55	34.62	25.48
Matsuzaki and Itaki (2017)	KR09-10 PC01-01 A	140.49	34.72	25.90
Matsuzaki and Itaki (2017)	GH80-2-409	141.02	34.89	25.71
This study	KH-22-7-OP-MC12 0–1 cm	154.97	34.98	25.16
Matsuzaki and Itaki (2017)	GH80-2-407	140.23	35.01	25.79
Matsuzaki and Itaki (2017)	GH80-2-411	141.09	35.15	25.53
This study	KS-24-6-MC01-H-0–1 cm	141.77	36.10	23.61
Matsuzaki and Itaki (2017)	MR01-ENG MC01-H01 W	143.67	36.42	23.75
Matsuzaki and Itaki (2017)	GH81-2-271	141.66	36.67	23.61
Matsuzaki and Itaki (2017)	GH81-2-174	141.67	37.40	21.22
This study	YK21-07S-PL06 0–2 cm	147.50	37.45	23.95
This study	KH-22-7-OP-MC13 0–1 cm	155.00	37.52	23.45
Matsuzaki and Itaki (2017)	MR98-05 MC06 (St.5)-H**W	162.72	37.76	21.32
Matsuzaki and Itaki (2017)	GH81-2-113	142.09	37.89	21.17
Matsuzaki and Itaki (2017)	GH81-2-71	142.17	38.07	21.05
Matsuzaki and Itaki (2017)	GH81-2-44	141.84	38.20	20.90
Matsuzaki and Itaki (2017)	GH81-2-30	142.17	38.27	20.85
Matsuzaki and Itaki (2017)	GH81-1-86	142.12	38.81	20.20
Matsuzaki and Itaki (2017)	GH81-1-69	142.42	38.91	20.08
Matsuzaki and Itaki (2017)	GH81-1-38	142.17	39.20	19.74
Matsuzaki and Itaki (2017)	GH81-1-30	142.09	39.30	19.63

Table 1 (continued)

Reference	Samples	Longitude (°E)	Latitude (°N)	Summer sea surface temperature (°C)
Matsuzaki and Itaki (2017)	GH81-1–13	142.42	39.50	19.39
This study	KH-22–7-OP-MC14 0–1 cm	155.03	39.93	20.41
Matsuzaki and Itaki (2017)	MR99-K04 MC02-H*1 W	149.85	40.08	19.25
Matsuzaki and Itaki (2017)	KR08-10 PC03-01 A	147.71	40.33	18.46
Matsuzaki and Itaki (2017)	MR98-05 MC01 (St.1)-H**W	142.99	40.51	18.20
Matsuzaki and Itaki (2017)	GH82-3–13	141.79	40.94	17.73
Matsuzaki and Itaki (2017)	GH82-3–16	141.92	40.97	17.72
This study	KH-22–7-OP-MC18 0–1 cm	149.98	40.98	18.42
Matsuzaki and Itaki (2017)	GH82-3–57	141.92	41.41	17.50
Matsuzaki and Itaki (2017)	GH03-282	143.17	41.62	16.88
Matsuzaki and Itaki (2017)	GH82-3–111	141.67	41.77	17.60
Matsuzaki and Itaki (2017)	GH03-200	144.01	41.93	16.58
This study	KS-24–6-MC03-H-0–1 cm	144.85	42.15	16.24
Matsuzaki and Itaki (2017)	GH03-102	144.07	42.37	16.56
Matsuzaki and Itaki (2017)	GH03-109	144.50	42.53	13.76
This study	KH-22–7-OP-MC15 0–1 cm	155.00	44.00	14.25
Matsuzaki and Itaki (2017)	GH00-151	144.07	44.40	12.42
Matsuzaki and Itaki (2017)	GH00-150	144.07	44.47	12.42
Matsuzaki and Itaki (2017)	GH00-149	144.07	44.53	12.40
Matsuzaki and Itaki (2017)	GH00-148	144.07	44.60	12.36
Matsuzaki and Itaki (2017)	GH00-147	144.07	44.67	12.32
Matsuzaki and Itaki (2017)	GH00-146	144.07	44.73	12.29
Matsuzaki and Itaki (2017)	GH00-145	144.07	44.80	12.26
Matsuzaki and Itaki (2017)	GH00-143	144.07	44.93	12.21
Matsuzaki and Itaki (2017)	GH00-142	144.07	45.00	12.19
Matsuzaki and Itaki (2017)	MR01-K03 MC03-H04 W	167.51	46.39	10.56
Matsuzaki and Itaki (2017)	MR00-K03 MC04-H06 A	154.78	49.37	7.54
Motoyama et al. (2016)	13	130.06	34.70	25.47
Motoyama et al. (2016)	366	130.76	35.75	24.35
Motoyama et al. (2016)	44	135.00	36.00	24.93
Motoyama et al. (2016)	267	131.70	36.10	24.03
Motoyama et al. (2016)	98	135.43	36.16	24.99
Motoyama et al. (2016)	48	135.04	36.29	24.99
Motoyama et al. (2016)	100	135.45	36.30	24.99
Motoyama et al. (2016)	236	134.30	36.34	24.99
Motoyama et al. (2016)	P104	131.12	36.40	24.03
Motoyama et al. (2016)	102	135.47	36.43	24.98
Motoyama et al. (2016)	51	135.07	36.51	24.98
Motoyama et al. (2016)	104	136.59	36.59	24.39
Motoyama et al. (2016)	150	135.78	36.72	24.98
Motoyama et al. (2016)	215	136.33	36.80	24.98
Motoyama et al. (2016)	217	136.20	36.90	24.98
Motoyama et al. (2016)	P103	130.55	36.92	24.91
Motoyama et al. (2016)	223	135.80	37.20	24.39
Motoyama et al. (2016)	139	136.41	37.29	24.91
Motoyama et al. (2016)	24	135.32	37.46	24.13
Motoyama et al. (2016)	27	135.92	37.56	24.13
Motoyama et al. (2016)	28	136.01	37.58	24.39

Table 1 (continued)

Reference	Samples	Longitude (°E)	Latitude (°N)	Summer sea surface temperature (°C)
Motoyama et al. (2016)	P106	131.55	37.59	23.49
Motoyama et al. (2016)	29	136.10	37.61	24.91
Motoyama et al. (2016)	32	138.14	37.63	24.91
Motoyama et al. (2016)	13	135.96	37.69	24.39
Motoyama et al. (2016)	103	138.27	38.29	24.53
Motoyama et al. (2016)	701	136.18	38.35	23.84
Motoyama et al. (2016)	120	138.21	38.42	24.53
Motoyama et al. (2016)	P127	135.28	38.52	23.49
Motoyama et al. (2016)	139	138.16	38.56	24.53
Motoyama et al. (2016)	B188	138.95	38.57	24.53
Motoyama et al. (2016)	161	138.10	38.69	24.53
Motoyama et al. (2016)	182	138.05	38.82	24.53
Motoyama et al. (2016)	204	137.99	38.95	23.42
Motoyama et al. (2016)	288	139.23	39.17	23.42
Motoyama et al. (2016)	731	138.23	39.23	23.42
Motoyama et al. (2016)	709	137.46	39.35	23.07
Motoyama et al. (2016)	B343	139.84	39.45	23.42
Motoyama et al. (2016)	703	136.50	39.50	22.70
Motoyama et al. (2016)	P125	133.07	39.69	21.94
Motoyama et al. (2016)	710	137.26	39.70	23.07
Motoyama et al. (2016)	711	137.70	39.74	23.07
Motoyama et al. (2016)	717	139.31	40.43	22.72
Motoyama et al. (2016)	P129	136.93	40.74	20.17
Motoyama et al. (2016)	727	139.01	40.88	21.88
Motoyama et al. (2016)	16	139.75	41.40	21.88
Motoyama et al. (2016)	10	138.75	41.40	21.29
Motoyama et al. (2016)	1203	139.19	41.83	21.88
Motoyama et al. (2016)	1101	138.02	41.99	21.29
Motoyama et al. (2016)	90	139.82	42.00	21.88
Motoyama et al. (2016)	143	138.94	42.58	20.67
Motoyama et al. (2016)	153	139.51	42.67	20.67
Motoyama et al. (2016)	P111	137.77	43.73	18.60
Motoyama et al. (2016)	1208	138.83	43.77	19.34
Motoyama et al. (2016)	334	140.92	43.92	20.28
Motoyama et al. (2016)	279	140.25	43.94	20.28
Motoyama et al. (2016)	P110	138.39	45.02	16.95

during winter is minimal (Da Silva et al. 1994). The subarctic gyre exerts an influence in the middle-to-high latitudes, primarily through the OC, which circulates counterclockwise and transports cold water from the Bering Sea into the northwestern Pacific (Tomczak and Godfrey, 1994). This leads to cooling of the water masses in the region, with summer SSTs dropping to around 12 °C, at latitudes above 42°N (Fig. 1). In addition, this region is impacted by the North Pacific Intermediate Water (NPIW), a cold water mass of moderately low

salinity that originates from brine rejection in the Sea of Okhotsk (Talley 1993). This rejection leads to the formation of the Okhotsk Sea Intermediate Water (OSIW), the coldest and freshest water mass in the area, which contributes to the NPIW's formation and carries low-temperature water into the lower layers of the North Pacific (Yasuda et al., 1997; 2003). Close to the area where the NPIW forms area, water temperatures at intermediate depths (500–700 m) remain at around 3 °C.

The Sea of Japan is a semi-enclosed marginal sea located in the middle latitudes of the northwestern Pacific Ocean and is connected to the northwestern Pacific through straits with depths of less than 130 m; it is thus isolated from the intermediate and deep waters of the North Pacific (Gamo et al. 2014) (Fig. 1). The only currents flowing into the Sea of Japan are the Tsushima Warm Current, which transports heat, freshwater, and nutrients from the East China Sea and is influenced by the East Asia summer monsoon and the Kuroshio Current. These currents thus control the marine environment and ecosystem of the Sea of Japan (Gamo et al. 2014). During the East Asia winter monsoon, the cooling of the surface water masses in the northern part of the Sea of Japan leads to the formation of deep waters through downwelling of the surface waters and to dissolved oxygen being supplied to the sea's intermediate and deep waters (Gamo et al. 2014). These intermediate and deep waters, known as the Japan Sea Proper Water (JSPW), have high dissolved oxygen concentrations, high biodiversity, and very low temperatures. These localized water masses have a residence time of about 100 years (Gamo et al. 2014).

2.2 New material and revisited methodology

2.2.1 New materials treatments and new compilation of a Northwest Pacific dataset

Prior the review, radiolarian microscopic slides were mounted from the newly collected 33 surface sediments collected from East China Sea and the Central Northwest Pacific during Japan Joint Usage/Research Center for Atmosphere and Ocean Science Expedition KH-21-3, YK-21-07, KH-22-3, KS-22-4, KH-22-7 and KS-24-6 (Table 1). The samples were first dried in an oven at 40°C for 14 days, and then 4 g of dried samples were treated with diluted hydrogen peroxide (10%) and hydrochloric acid (5%) to remove all the organic and calcareous matter. The undissolved residue in each sample was sieved over a 45 µm screen and mounted onto a 22 × 40 mm cover glasses (microscopic slides) following the procedure of Matsuzaki and Itaki (2017). Examination of polycystine radiolarians was carried out under an optical microscope at magnifications of 100 × to 400 × following criteria of Itaki (2008); Matsuzaki et al. (2015a, b) and Zhang and Suzuki (2017). Observations of samples were continued until at least 400 radiolarians were identified in each sample, or until the sample material was exhausted (in average 427 radiolarians per sample). We calculated the relative abundances of each species from the total assemblages (Supplementary data).

Then, we integrated this new dataset of 33 samples with those proposed by Matsuzaki and Itaki (2017) (89) and Motoyama et al. (2016) (57), thus defining a

dataset of 179 samples that encompassed the Northwest Pacific from longitude 150°E to its Asian marginal seas, including the East China Sea and the Sea of Japan (Table 1; Fig. 1). This expanded dataset enabled us to investigate the geographical distribution patterns of radiolarian species and to develop updated SST estimation methods for the entire Northwest Pacific region.

2.2.2 International ocean drilling program expedition 346 site U1429

To apply our new methodology, we re-analyzed the radiolarian dataset of Matsuzaki et al. (2019), which was obtained at site U1429 in the ECS. This site is situated in the northernmost part of the ECS at a depth of 732 m at coordinates 31°37.04' N and 128°59.85' E (Tada et al. 2015). This location falls within the southern area of the Danjo Basin, part of the northern section of the Okinawa Trough (Tada et al. 2015). The sediment cores retrieved from this site are approximately 180 m long and are composed of calcareous nannofossil-rich clay containing diatoms and foraminifera (Tada et al. 2015). The sedimentary sequence has been divided into two lithologic units, classified as Unit A and Unit B. Unit A covers the Holocene to Middle Pleistocene period and consists of olive-gray to light-green calcareous-rich clay interspersed with tephra layers (Tada et al. 2015). In this study, we used the core composite depth below seafloor (CCSF-D) scale of Irino et al. (2018). The depths in CCSF-D were then converted to ages (ka, with 1 ka = 1000 years) following the age model proposed by Sagawa et al. (2018), which relies upon the oxygen isotope stratigraphy of benthic foraminifer tuned to the LR04 benthic stack (Lisiecki and Raymo 2005).

2.2.3 Species and water depths selection for sea surface temperature estimates

Matsuzaki and Itaki (2017) suggested that for accurate estimation of past sea surface temperatures (SSTs), selecting species inhabiting the upper 200 m of the water column was crucial. This zone generally remains warmer due to solar heating and mixing by wind, waves, and currents (Talley 1991); and sunlight penetration diminishes with depth, becoming insufficient for photosynthesis below 200 m (Talley 1991). Since most tropical radiolarian species conduct photosynthesis through algal symbiosis (Zhang et al. 2018), they are likely sensitive to photic zone conditions. Furthermore, in the East China Sea, a tropical marginal sea of the Northwest Pacific, radiolarian fluxes peaked between 0 and 25 m depth, with the second highest fluxes at 25–50 m (Matsuzaki et al. 2016). Therefore, in the tropics and areas influenced by warm water masses, radiolarians appear highly sensitive to

conditions within the upper 25 m of the water column. However, in subarctic regions, where diatoms and sea ice often dominate the surface, radiolarians primarily inhabit the water depths between 50 and 200 m (Okazaki et al. 2004).

Therefore, radiolarians are closely related to the conditions of the upper 200 m of the water column in the Northwest Pacific. However, the key point is that depending on whether the area is influenced by warm water masses or not, the radiolarian assemblages infer information from different water masses. For example, in the tropics or areas influenced by the Kuroshio Current and/or its bifurcated current, radiolarians are likely related to the surface conditions (0–25 m). Thus, it would be fair to assume radiolarian SSTs infer temperatures at 0 m water depth. In contrast, in the subarctic area, radiolarians more likely infer conditions at ~100 m. Because a wide area of the samples we re-analyzed in this study are influenced by the Kuroshio Current system, we assume in the context of this study that radiolarian-based SSTs infer those at 0 m water depth, acknowledging that they may infer SSTs at deeper water depths in the subarctic areas.

Previous plankton tow studies have highlighted that the following 30 species have been consistently found to inhabit the upper 200 m of the water column in the Northwest Pacific, including the Sea of Japan and the East China Sea, and have the potential to be used for SST reconstructions (e.g., Itaki 2003; Okazaki et al. 2004; Itaki et al. 2010; Ishitani et al. 2008; Matsuzaki et al. 2016, 2020a). This group consists of the species *Spongodiscus resurgens*, *Stylochlamyidium venustum*, *Tholospira buetschlii*, *Lychnospaera boreale*, *Antarctissa* (?) sp.1., *Ceratospyris borealis*, *Dictyocoryne profunda/truncatum*, *Dictyocoryne muelleri*, *Didymocyrtis tetrathalamus*, *Disolenia* spp., *Heliodiscus asteriscus/echniscus*, *Heliodiscus macrococcus*, *Phortidium* spp., *Polysolenia lappacea/spinosa*, *Spongaster tetras tetras*, *Spongosphaera streptacantha*, *Lithomelissa setosa*, *Botryocyrtis scutum*, *Rhizospaera medianum*, *Acanthodesmia vinculata*, *Dictyocryphalus hispida*, *Dictyopodium praetextum*, *Pseudodictyophimus gracilipes*, *Tetrapyle fruticosa/circularis*, *Zygocircus* spp., *Stylodictya tenuispina/stellata*, *Collosphaera tuberosa*, *Lampromitra erosa*,

Botryostrobilus auritus, and *Archipерidium calvata*. The names of the genus were updated following Suzuki et al. (2021) and O'Dogherty et al. (2021). In this study, we propose to re-analyze these species' affinities with changes in SSTs through several statistical analyses and select species that are well constrained by temperature changes for further SST estimates. The key species used later for SST estimates are illustrated in Fig. 2.

2.2.4 Statistical re-analysis of the new compiled dataset

In this study, before checking the selected radiolarian species sensitivity to SST changes, we aimed to determine if the 30 selected species have a linear or unimodal response to ecological gradients in the Northwest Pacific, including the open ocean, the Japan Sea, and the East China Sea. To do this, we applied Detrended Correspondence Analysis (DCA) to the abundances of the selected species listed above based on our dataset of 179 samples obtained from the Northwest Pacific from longitude 150°E to the Asian Marginal Seas (Table 1; Fig. 1). Before conducting any statistical analysis, percentage data should satisfy the assumption of unimodal distribution. To satisfy this requirement, we applied a log-ratio transformation to the data to obtain a better normalized dataset prior to performing DCA. This transformation offers two advantages: it resolves the independence violation and produces better normalized data, making them more suitable for multivariate analyses (e.g., Kucera and Malmgren 1998). Following the log-ratio transformation of the 30 selected species relative abundances, we performed DCA using Past4 software (Hammer et al., 2001).

The DCA analysis ordination diagram displays the distribution of both species and samples along the two main axes (Axis 1 and Axis 2) (Fig. 3). The ordination shows that most samples and species are distributed along the first DCA axis (DCA1), whereas Japan Sea samples are arranged orthogonally to those from the East China Sea and Northwest Pacific (Fig. 3). This pattern suggests that the Japan Sea ecosystem is influenced by distinct environmental factors and may harbor a unique species assemblage compared to the neighboring regions, likely due to the unique oceanographic conditions of this marginal sea as discussed in 2.1. Given that our review aims

(See figure on next page.)

Fig. 2 1. *Polysolenia lappacea* (KH-22-7-OP11-MC-0-1 cm); 2. *Polysolenia spinosa* (KH-22-7-OP11-MC-0-1 cm); 3. *Stylochlamyidium venustum* (KH-22-7-OP18-MC-0-1 cm); 4. *Spongodiscus resurgens* (KH-22-7-OP18-MC-0-1 cm); 5. *Didymocyrtis tetrathalamus* (KS-22-4-MC04-0-1 cm); 6. *Lychnospaera boreale* (KS-24-6-MC03-0-1 cm); 7. *Stylodictya tenuispina* (KH-22-7-OP15-MC-0-1 cm); 8. *Tholospira buetschlii* (KH-22-7-OP15-MC-0-1 cm); 9. *Dictyocoryne profunda* (KH-22-7-OP04-MC-0-1 cm); 10. *Dictyocoryne truncatum* (KH-22-7-OP04-MC-0-1 cm); 11. *Dictyocoryne muelleri* (KH-22-7-OP04-MC-0-1 cm); 12–13. *Phortidium* spp. (KH-22-7-OP04-MC-0-1 cm); 14–15. *Tetrapyle fruticosa/circularis* group (KS-22-4-MC04-0-1 cm); 16. *Dictyopodium praetextum* (KS-22-4-MC04-0-1 cm); 17–18. *Zygocircus* spp. (KS-22-4-MC04-0-1 cm); 19. *Botryocyrtis scutum* (KS-22-4-MC04-0-1 cm); 20. *Lithomelissa setosa* (KS-24-6-MC03-0-1 cm); 21. *Ceratospyris borealis* (KH-22-7-OP15-MC-0-1 cm); 22. *Pseudodictyophimus gracilipes* (KH-22-7-OP18-MC-0-1 cm); 23. *Acanthodesmia vinculata* (KS-22-4-MC04-0-1 cm)

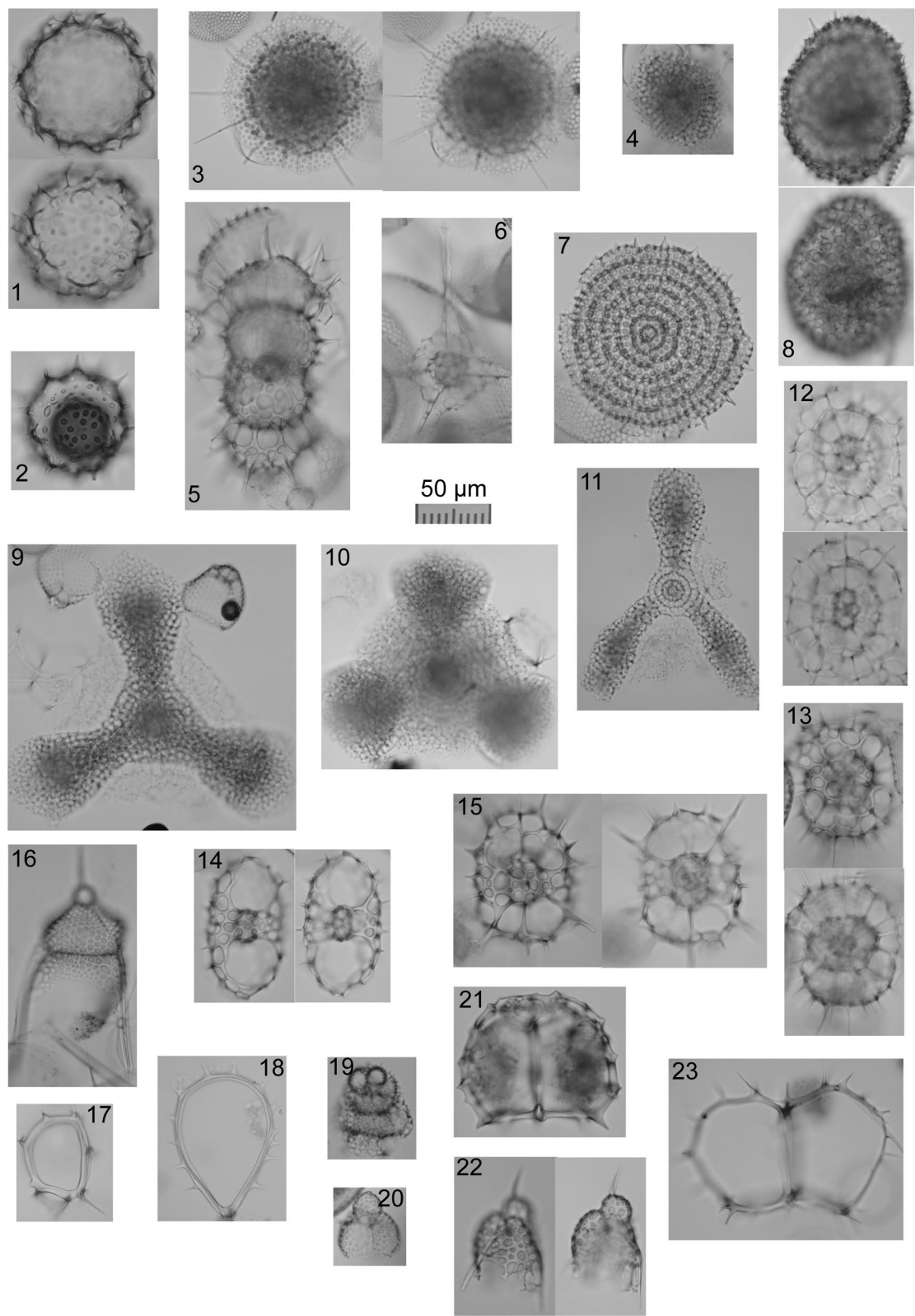


Fig. 2 (See legend on previous page.)

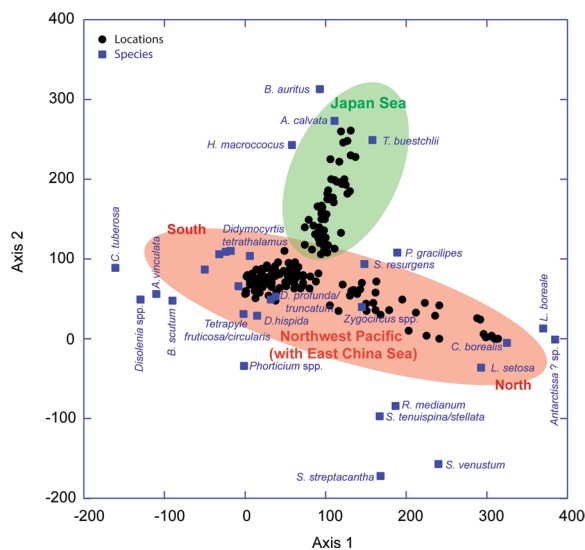


Fig. 3 Detrended Correspondence Analysis (DCA) was conducted on the core-top dataset obtained in this study. The two principal axes, Axis 1 and Axis 2, were cross-plotted in an XY plot. The black dots represent the collected sediment core tops, while the blue squares represent the distributions of key radiolarian species

to establish radiolarian species as suitable proxies for SST estimates in the Northwest Pacific, including Japan Sea samples could introduce bias and reduce the accuracy of SST reconstructions. Therefore, based on the DCA results, we excluded the 57 samples from the Japan Sea from the SST estimates and focused on the 122 samples from the East China Sea and the Northwest Pacific. These 122 surface sediment samples will be used for Factor Analysis to determine key radiolarian species affinities with water masses and SSTs in the following, while the Japan Sea assemblages will be assessed in a future study.

Factor Analysis was thus performed on 122 samples from the East China Sea and the Northwest Pacific, using log-transformed relative abundances of selected species from core-top data and summer SSTs at 0 m water depth (Locarnini et al. 2018) as the environmental variable. Previous North Pacific studies have suggested that core-top radiolarian assemblages represent annual mean conditions, favoring the use of annual mean SSTs for multivariate regression analyses (e.g., Kamikuri et al. 2008; Matsuzaki et al. 2014). In tropical and monsoon-affected regions, where phytoplankton blooms occur during winter monsoons, some researchers have proposed that radiolarian-based SSTs better reflect winter conditions (e.g., Hernández-Almeida et al. 2017; Matsuzaki et al. 2020b). However, studies conducted north of Japan have shown that radiolarian blooms produce approximately 700 specimens/m²/day during late summer and 1300 specimens/m²/day in winter (Itaki et al.

2008). While this might suggest winter SSTs are more appropriate, Itaki et al. (2008) also reveals that a single species, *L. setosa*, accounts for 800 specimens/m²/day of the winter flux and *L. setosa* is primarily confined to northern Japanese coastal waters in modern condition. Excluding *L. setosa*, winter radiolarian fluxes drop below 500 specimens/m²/day, which is less than the summer peak of 700 specimens/m²/day, which comprises species commonly found in the subtropical North Pacific gyres such as *Zygocircus* spp., *D. tetrathalamus*, and *T. circularis/fruticosa* (Itaki et al. 2008).

While our literature review suggests summer SST at 0 m as the most suitable variable, we recognize the need for statistical verification. The DCA results showed that excepting the Japan Sea, samples spread less than 4 standard deviations along axis 1 (Fig. 3), indicating that species responses to environmental gradients were predominantly linear rather than unimodal. Therefore, we conducted a Redundancy Analysis (RDA) using PAST4 software to assess the relative importance of winter and summer SST on radiolarian species distribution. We utilized the same log-normalized dataset used for DCA and intended for following Factor Analysis. The RDA results show that certain species, such as *C. tuberosa*, *Dissolenia* spp., *P. lappacea/spinosa*, *Zygocircus* spp., the *D. profunda/truncatum* group, and *S. tetras tetras*, are more influenced by winter SST (Fig. 4). Conversely, other species, including the *Tetrapyle fruticosa/circularis* group, *Phortidium* spp., *D. praetextum*, *D. hispida*, the *S. tenuispina/stellata* group, *A. calvata*, *D. muelleri*, and *D. tetrathalamus*, appear more influenced by summer SST (Fig. 4). This finding complicates the decision. However, within the surface sediments of our study area, relative abundances of winter SST-associated species rarely exceed 3% each. In contrast, summer SST-associated species, such as the *Tetrapyle* group, *Phortidium* spp., *D. tetrathalamus*, and *D. muelleri*, frequently exhibit high abundances, reaching 10% to 30% each (see appendix). Interestingly, species associated with subarctic regions show no significant correlation with either winter or summer SST (Fig. 4). Considering that more than half of the total radiolarian assemblages of most locations of the study area are comprised of species associated with summer SST, as suggested by the RDA results, and supported by our previous literature review, we selected summer SSTs at 0 m water depth (Locarnini et al. 2018) as the variable for our Factor Analysis to investigate species affinities with temperature changes.

After reviewing, statistics and confirming key species affinities with SST changes, we selected species showing strong relationships with Summer SST at 0 m water depths and applied a weighted averaging partial least squares (WA-PLS) analysis (ter Braak and Juggins 1993)

Table 2 Factor Analysis results with its 10 factors and their respective eigenvalues, variance, and cumulative variance

Factors	Eigenvalue	% variance	Cumulative (%)
1	83.18	68.18	68.18
2	17.65	14.47	82.65
3	5.72	4.69	87.34
4	2.20	1.81	89.15
5	1.99	1.64	90.79
6	1.23	1.01	91.80
7	1.16	0.95	92.75
8	1.08	0.88	93.63
9	0.90	0.74	94.37
10	0.75	0.62	94.99

to reconstruct SSTs between 400 and 0 ka interval at Site U1429 in the northern East China Sea using PAST4 software (Hammer et al., 2001). When applying WA-PLS, the optimal number of components should be selected to obtain the lowest Root Mean Square Error of Prediction (RMSEP) (ter Braak and Juggins 1993). In this study, the analysis using two components provided the lowest root-mean-square error of prediction (RMSEP) of 1.41 °C with a high coefficient of determination (R^2) of 0.97. Last, we assessed the dissimilarity between fossil and modern assemblages by calculating the Squared Chord Distance (SCD) using Past4 (Hammer et al., 2001).

2.3 Shallow-water radiolarian species distributions in the Northwest Pacific and their affinity with oceanography (water masses and currents)

In the previous section, DCA demonstrated differences between radiolarian assemblages from the Sea of Japan and Northwest Pacific/East China Sea (ECS) samples. In this section, we examine the relationships between selected 30 species and sea surface temperature (SST) in the Northwest Pacific and the ECS by applying Factor Analysis to the radiolarian dataset and summer SSTs at 0 m water depth.

Factor Analysis identified four factors that explained more than 89% of the dataset variance (Table 2). Factor 1 was the most significant, accounting for 68.18% of the variance, followed by Factor 2 (14.47%), Factor 3 (4.69%), and Factor 4 (1.81%) (Table 2). To better visualize the spatial distribution patterns, we mapped the factor loadings (Table 3; Fig. 5) using Ocean Data View 4 (ODV4) (Schlitzer et al., 2022). Statistically, for species, factor scores above 1.0 indicate that a species contributes substantially to the environmental patterns being analyzed (SST ranges in this study).

2.3.1 Subtropical assemblages

Factor 1 exhibits high loadings (> 0.5) from the equator to 35°N (Table 3; Fig. 5). These high loadings correspond to regions with summer SSTs exceeding 25 °C (Table 3; Fig. 6). Species with factor scores higher than 1 include: *T. circularis/fruticosa* group (3.06), *Zygocircus* spp. (1.85), *D. tetrathalamus* (1.74), *Phortidium* spp. (1.79), *D. profunda/truncatum* (1.69), *P. lappacea/spinosa* (1.1), *D. muelleri* (1.09), *B. scutum* (1.01), *A. vinculata* (1.06), and *D. praetextum* (1.0) (Table 4). As these species are all documented as warm water indicators (e.g., Lombardi and Boden, 1985; Matsuzaki and Itaki 2017), we interpret Factor 1 assemblages as subtropical assemblages associated with SSTs higher than 25 °C.

A further mapping was conducted using ODV4 to visualize the relative abundances of species that showed highest factor scores for Factor 1 (Fig. 7). The distribution patterns reveal that *T. circularis/fruticosa* group reaches its highest abundances (25–30%) in the East China Sea, maintaining abundances above 15% between the equator and 35°N. *Phortidium* spp. shows more moderate abundances, with maxima of approximately 12% in the East China Sea and between 30°N and 40°N. *Zygocircus* spp. shows notable concentrations in the Northwest Pacific Warm Pool region (10°N–20°N). *D. profunda/truncatum* display similar distribution patterns, with higher abundances in the equatorial region. *D. tetrathalamus* shows high abundances in areas directly influenced by the Kuroshio Current and its source region, including the Philippine Sea and the West Pacific Warm Pool (Fig. 7).

Species commonly used in palaeoceanography, such as *T. circularis/fruticosa* and *D. tetrathalamus*, appear to be associated with the warm, saline, shallow Kuroshio Current (KC) waters (Fig. 6). However, as noted by Qu et al. (2022), these species have different ecological preferences, with *T. circularis/fruticosa* showing maximum relative abundances in the East China Sea (ECS) (Fig. 7). The ECS is influenced by both the KC and freshwater discharge from the Changjiang River in the northwestern region—known as the “Changjiang Diluted Water” (CDW) (Ichikawa and Beardsley 2002). This freshwater discharge dominates the shallow continental shelf waters in the northern ECS before mixing with the warmer, more saline KC waters, creating highly productive, nutrient-enriched shallow waters (Ichikawa and Beardsley 2002). Therefore, while *T. circularis/fruticosa* is associated with the KC, its abundance appears enhanced in the ECS due to CDW influence. Studies from the South China Sea by Hernández-Almeida et al. (2017) and Qu et al. (2020; 2022) have also linked the *Tetrapyle* group to eutrophic upwelling zones in tropical seas. Therefore, *T. circularis/fruticosa* has a strong affinity for warm waters in the Northwest Pacific; however,

Table 3 List of samples, their associated summer SSTs at 0 m water depth (Locarnini et al. 2018), and their factor loadings obtained from Factor Analysis conducted on selected 30 taxa

Samples	Summer sea surface temperature (°C)	Factor 1	Factor 2	Factor 3	Factor 4
KH-22-7-OP-MC03 0–1 cm	29.31	0.92	−0.08	0.20	0.09
KH-22-7-OP-MC02 0–1 cm	29.36	0.95	−0.08	0.19	0.07
NH90-1-N54-NB17	29.12	0.92	−0.03	0.22	0.15
NH90-1-N56-NB18	29.171	0.89	−0.06	0.14	0.16
KH-22-7-OP-MC04 0–1 cm	29.39	0.93	−0.09	0.16	−0.04
NH90-1-N57-NB19	29.222	0.92	−0.07	0.18	0.08
NH90-1-N60-NB21	29.249	0.93	−0.08	0.16	0.16
NH90-1-N62-NB22	29.234	0.94	−0.09	0.11	0.10
KH-22-7-OP-MC05 0–1 cm	29.29	0.96	−0.09	0.15	0.09
NH90-1-N68-NB25	29.238	0.96	−0.08	0.09	0.05
KH-22-7-OP-MC01 0–1 cm	29.44	0.95	−0.09	0.12	0.06
KH-22-7-OP-MC070–1 cm	29.21	0.89	−0.11	0.12	0.02
KR02-01 PC03-01 A	29.066	0.94	−0.06	0.14	−0.06
KR13-02 PC03-01 W	28.738	0.92	−0.10	0.09	0.11
NH90-1-N79-NB30	28.518	0.94	−0.05	0.25	0.08
KH-21-3 PL04 0–2 cm	28.85	0.96	−0.07	0.17	−0.06
KH-22-7-OP-MC08 0–1 cm	28.73	0.94	−0.08	0.18	−0.04
KH-21-3 PL03 0–2 cm	28.85	0.90	−0.12	0.12	0.02
KH-21-3 PL02 0–2 cm	28.85	0.94	−0.08	0.20	−0.04
KH-21-3 PL01 0–2 cm	28.85	0.89	−0.11	0.17	0.06
GH08-198	28.492	0.93	−0.07	0.27	0.08
GH08-214	28.483	0.95	−0.08	0.08	−0.02
KH-22-3-PL08-0–2 cm	28.7	0.91	−0.05	0.20	−0.09
GH08-197	28.466	0.92	−0.05	0.25	0.08
GH08-247	28.462	0.96	−0.06	0.16	0.01
GH08-248	28.445	0.94	−0.05	0.16	0.03
GK14-612	28.348	0.94	−0.04	0.17	0.01
GK14-642	28.292	0.95	−0.03	0.25	−0.05
GK14-694	28.309	0.94	−0.05	0.21	−0.03
GK14-591	28.198	0.93	−0.03	0.23	−0.06
GK14-766	28.216	0.94	−0.02	0.15	−0.04
KS-22-4 MC04 0–1 cm	28.59	0.88	−0.06	0.21	−0.22
KS-22-4 MC07 0–1 cm	28.59	0.94	−0.02	0.22	−0.07
KH-22-3-PL01-0–2 cm	28.68	0.92	−0.06	0.17	−0.09
KH-22-7-OP-MC09 0–1 cm	27.33	0.82	0.02	0.36	−0.20
GH84-3-63	27.925	0.85	−0.11	0.35	−0.19
GH84-3-169	27.881	0.89	−0.04	0.35	−0.11
GH84-3-30	27.894	0.89	−0.10	0.31	−0.19
GH84-3-169	27.896	0.89	0.00	0.29	−0.05
GH84-3-37	27.872	0.88	−0.03	0.35	−0.16
GH84-3-263	27.662	0.90	−0.06	0.32	−0.16
GH84-3-224	27.715	0.87	0.01	0.39	−0.10
GH84-3-34	27.856	0.88	−0.05	0.36	−0.04
KH-21-3 MC06 0–1 cm	27.46	0.84	−0.07	0.34	−0.31
KH-21-3 MC04 0–1 cm	27.46	0.82	−0.02	0.32	−0.26
KH-21-3 MC03 0–1 cm	27.46	0.80	−0.10	0.34	−0.29
KH-21-3 MC01 0–1 cm	27.46	0.84	−0.09	0.40	−0.30
KH-21-3 MC02 0–1 cm	27.46	0.68	−0.07	0.48	−0.44

Table 3 (continued)

Samples	Summer sea surface temperature (°C)	Factor 1	Factor 2	Factor 3	Factor 4
KH-21-3 MC05 0–1 cm	27.46	0.80	−0.12	0.42	−0.27
GH83-2-320	27.633	0.87	−0.02	0.32	−0.23
GH83-2-264	27.633	0.82	−0.04	0.40	−0.30
IODP-346-U1429 A-1H1-27-29	27.06	0.65	−0.16	0.55	−0.23
KH-22-7-OP-MC10 0–1 cm	26.36	0.83	−0.03	0.35	−0.23
GH83-2-158	27.556	0.85	−0.06	0.31	−0.29
GH83-2-109	27.519	0.86	0.04	0.34	−0.17
KH-22-7-OP-MC11 0–1 cm	26.3	0.83	−0.02	0.39	−0.25
KH-22-7-OP-MC21 0–1 cm	26.71	0.86	0.00	0.37	−0.17
GH82-1-24	27.503	0.70	0.04	0.44	−0.27
GH82-1-1	27.486	0.83	0.03	0.43	−0.06
GH82-1-49	27.498	0.80	−0.06	0.22	−0.34
GH82-1-51	27.486	0.89	−0.06	0.36	−0.10
GH82-1-69	27.498	0.76	0.00	0.50	−0.14
GH82-1-106	27.504	0.82	−0.11	0.46	−0.25
GH82-2-110	26.29	0.72	−0.26	0.40	−0.36
GH82-2-49	26.284	0.79	−0.19	0.39	−0.15
GH82-2-66	26.281	0.68	−0.13	0.54	−0.41
GH97-258	25.83	0.78	−0.11	0.50	−0.24
GH82-2-11	26.29	0.82	0.01	0.35	−0.23
GH82-2-32	26.285	0.73	−0.11	0.34	−0.17
GH97-118	26.15	0.73	−0.16	0.59	−0.18
GH97-126	26.353	0.69	−0.13	0.56	−0.36
GH97-135	26.347	0.66	0.04	0.53	−0.24
GH97-163	26.327	0.76	−0.04	0.55	−0.13
GH97-175	26.321	0.68	−0.22	0.54	−0.38
GH97-80	25.477	0.75	−0.02	0.54	−0.21
KR09-10 PC01-01 A	25.902	0.72	−0.08	0.58	−0.22
GH80-2-409	25.714	0.79	−0.07	0.53	−0.16
KH-22-7-OP-MC12 0–1 cm	25.16	0.75	0.07	0.51	−0.18
GH80-2-407	25.789	0.76	−0.11	0.51	−0.25
GH80-2-411	25.525	0.71	−0.17	0.59	−0.24
KS-24-6-MC01-H-0-1 cm	23.61	0.73	−0.04	0.47	−0.42
MR01-ENG MC01-H01 W	23.75	0.84	−0.05	0.29	−0.20
GH81-2-271	23.61	0.49	−0.23	0.79	−0.10
GH81-2-174	21.219	0.45	−0.23	0.79	−0.09
YK21-07S-PL06 0–2 cm	23.95	0.69	−0.13	0.54	−0.22
KH-22-7-OP-MC13 0–1 cm	23.45	0.78	0.01	0.46	−0.14
MR98-05 MC06 (St.5)-H**W	21.321	0.65	0.08	0.67	−0.10
GH81-2-113	21.167	0.38	−0.29	0.83	−0.12
GH81-2-71	21.046	0.34	−0.21	0.83	−0.18
GH81-2-44	20.901	0.35	−0.18	0.84	−0.04
GH81-2-30	20.846	0.26	−0.24	0.89	−0.07
GH81-1-86	20.2	0.37	−0.33	0.82	0.02
GH81-1-69	20.081	0.19	−0.24	0.87	−0.04
GH81-1-38	19.742	0.35	−0.22	0.84	−0.09
GH81-1-30	19.625	0.38	−0.34	0.77	−0.04
GH81-1-13	19.392	0.26	−0.33	0.85	0.01
KH-22-7-OP-MC14 0–1 cm	20.41	0.58	−0.08	0.62	−0.31

Table 3 (continued)

Samples	Summer sea surface temperature (°C)	Factor 1	Factor 2	Factor 3	Factor 4
MR99-K04 MC02-H*1 W	19.247	0.46	0.01	0.77	−0.07
KR08-10 PC03-01 A	18.462	0.63	−0.14	0.64	−0.32
MR98-05 MC01 (St.1)-H**W	18.201	0.23	−0.32	0.87	0.14
GH82-3-13	17.731	0.31	−0.38	0.80	−0.09
GH82-3-16	17.72	0.32	−0.41	0.75	−0.07
KH-22-7-OP-MC18 0-1 cm	18.42	0.55	−0.07	0.69	−0.13
GH82-3-57	17.5	0.19	−0.59	0.70	−0.21
GH03-282	16.882	0.03	−0.81	0.38	−0.23
GH82-3-111	17.601	0.25	−0.52	0.73	−0.07
GH03-200	16.582	0.03	−0.52	0.75	0.13
KS-24-6-MC03-H-0-1 cm	16.24	0.16	−0.60	0.61	−0.15
GH03-102	16.564	0.05	−0.60	0.73	0.03
GH03-109	13.755	−0.01	−0.74	0.52	−0.01
KH-22-7-OP-MC15 0-1 cm	14.25	0.42	−0.16	0.69	−0.13
GH00-151	12.419	−0.02	−0.81	0.45	0.07
GH00-150	12.419	0.03	−0.93	0.18	0.02
GH00-149	12.398	0.00	−0.97	0.19	0.01
GH00-148	12.359	−0.07	−0.95	0.21	−0.06
GH00-147	12.324	−0.01	−0.98	0.18	0.00
GH00-146	12.291	−0.05	−0.96	0.23	−0.04
GH00-145	12.262	−0.01	−0.97	0.16	0.01
GH00-143	12.209	−0.05	−0.96	0.23	−0.03
GH00-142	12.187	0.04	−0.90	0.26	0.10
MR01-K03 MC03-H04 W	10.556	0.10	−0.31	0.78	0.27
MR00-K03 MC04-H06 A	7.539	0.05	−0.37	0.69	0.33

unlike other subtropical species, it particularly blooms in highly productive, nutrient-enriched warm waters where its optimal blooming conditions exist. *D. tetrathalamus* shows high abundances in regions directly influenced by the Kuroshio Current (KC) and its source region (Fig. 7). These findings align with previous studies (Lombardi and Boden 1985; Motoyama and Nishimura 2005; Kamikuri et al. 2008), confirming the reliability of our results. Like *T. circularis/fruticosa*, *D. tetrathalamus* can serve as a KC tracer. However, unlike *T. circularis/fruticosa*, *D. tetrathalamus* prefers oligotrophic waters (Ishitani et al. 2008; Qu et al. 2022). Recent work by Qu et al. (2022) demonstrated that *D. tetrathalamus* shows only weak correlation with nutrient levels, such as phosphate. Therefore, *D. tetrathalamus* is likely associated with the warm, oligotrophic KC waters of the southern subtropical gyre, while *T. circularis/fruticosa* shows stronger affinity for the eutrophic KC waters of the northern subtropical gyre.

Phorticum spp. shows abundance peaks of approximately 12% between 30°N and 40°N in the Northwest Pacific (Fig. 7). Previously Matsuzaki et al. (2018) and

Zhang et al. (2018) have suggested that *Phorticum* spp. is more closely associated with transitional water masses than subtropical ones. Our results indicate that *Phorticum* spp. has higher relative abundances from a little south the region where the KC converges with the OC to the KC/OC mixing area (Fig. 7). In the KC/OC zone, KC and OC waters undergo vertical mixing due to density differences, resulting in decreased sea surface temperatures (SSTs) and enhanced nutrient content, as evidenced by chlorophyll *a* measurements off the Japanese coast (Talley et al. 1995; Itoh et al. 2015). Based on these observations, we conclude that *Phorticum* spp. can be considered an indicator of transitional warm water masses.

Species such as *D. profunda/truncatum* and *Zygocircus* spp. are primarily distributed from the equator to the West Pacific Warm Pool, a region characterized by the world highest sea surface temperatures (SSTs) (Locarnini et al. 2018), demonstrating a clear association between these species and high SSTs. However, Qu et al. (2022) recently demonstrated that these species

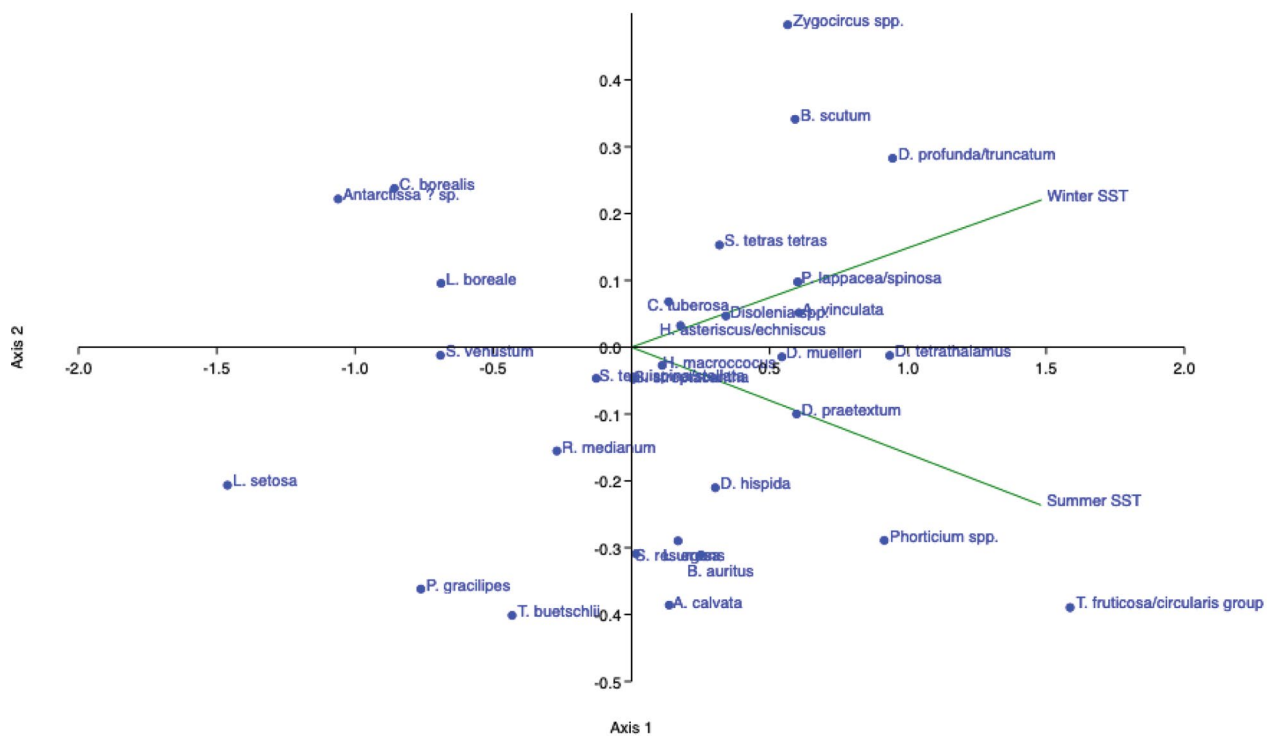


Fig. 4 A Redundancy Analysis (RDA) was conducted on the core-top dataset obtained in this study to assess the relationship between radiolarian species composition and summer and winter sea surface temperatures (SST, °C). The first two principal axes (Axis 1 and Axis 2) were visualized in a biplot. Blue dots represent the distribution of key radiolarian species, while green lines represent gradients of winter and summer SST

distributions may also be influenced by net primary production, phosphate levels, and food availability.

2.3.2 The sea of Okhotsk and surrounding subarctic Northwest Pacific assemblages

Factor 2 loadings are high (< -0.5) in samples between 40°N and 55°N, with the strongest loadings concentrated in the Sea of Okhotsk (Fig. 5) and corresponding to summer SSTs below 14 °C (Table 3; Figs. 5 and 6). The species *L. setosa*, *Antarctissa*? sp.1, *L. boreale*, and *C. borealis* has factor score of -3.5 , -2.6 , -1.85 , and -1.8 (Table 4), respectively, and were thus assumed to be associated with a Sea of Okhotsk and surrounding subarctic Northwest Pacific assemblages with summer SSTs lower than 14 °C.

L. setosa is the dominant species in the subarctic assemblage (Table 4) and exhibits high relative abundances, up to 20% off the Sea of Okhotsk and Pacific sides of Hokkaido (Fig. 8). As shown in Fig. 8, this species does not occur offshore, suggesting a connection to subarctic coastal waters. *L. setosa* is also commonly found in the eastern region of the Norwegian Sea, near the coast of Norway (Bjørklund et al., 1998). Research based on sediment traps in the areas adjacent to Hokkaido has indicated a correlation between *L. setosa* and a large influx of lithogenic material into the ocean during winter (Itaki

et al. 2008). In general, when a large amount of lithogenic matter enters the ocean, it leads to an increase in nutrients and biological productivity (e.g., Ruddiman 2001). This suggests that *L. setosa* is likely linked to higher productivity in subarctic shallow-water environments during winter. Thus, *L. setosa* can be related to the East Asian Winter Monsoon. In their study along the northwestern Pacific coast from the Philippines to Japan, Matsuzaki and Itaki (2017) demonstrated that *L. setosa* abundance increases significantly in colder coastal environments, particularly along northern Japan. In contrast, the *Tetrapyle* group dominates in both eutrophic subtropical and tropical coastal waters, showing a marked decrease in abundance northward. These contrasting distribution patterns in both species make *L. setosa* as an effective ecological counterpart to the *Tetrapyle* group for SST reconstructions.

Subarctic species, such as *C. borealis*, are found to be abundant in the Sea of Okhotsk, accounting for approximately 8% and 10% of the assemblage, respectively (Fig. 8). This species is also common in the subarctic Northwest Pacific, where its abundance is about 5% (Fig. 8). *L. boreale*, another species, has a high relative abundance of 6% in the Sea of Okhotsk but only about 2% in the subarctic Northwest Pacific (Fig. 8). Similar

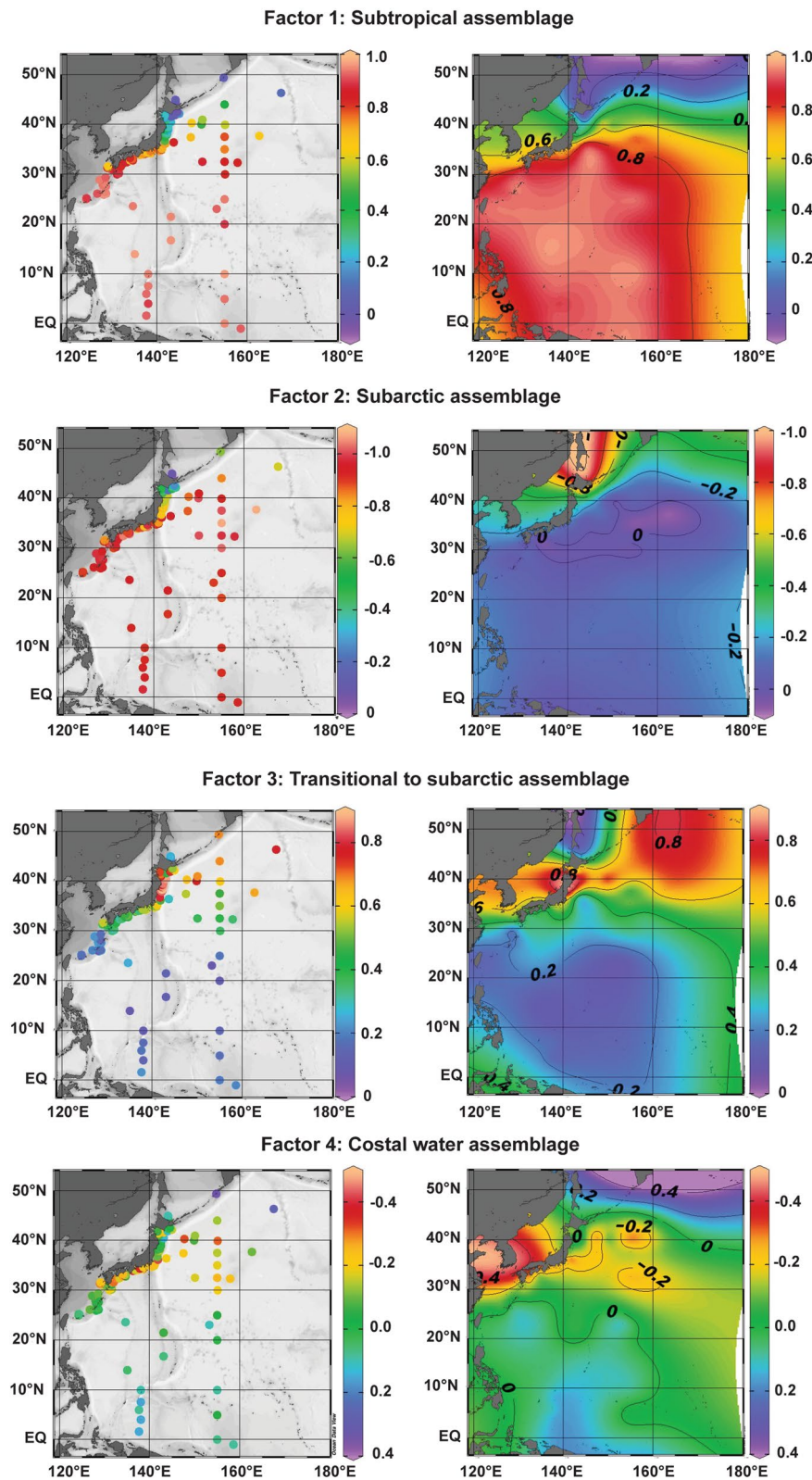


Fig. 5 Mapping of the four main principal factors (F1 to F4) obtained from a Factor Analysis conducted on the core-top dataset of this study. The maps were generated using ODV4 (Schlitzer et al., 2022). The left panels show raw loadings data, and the right panels show raw loadings data using DIVA (Data-Interpolating Variational Analysis) gridding interpolation (Schlitzer et al., 2022)

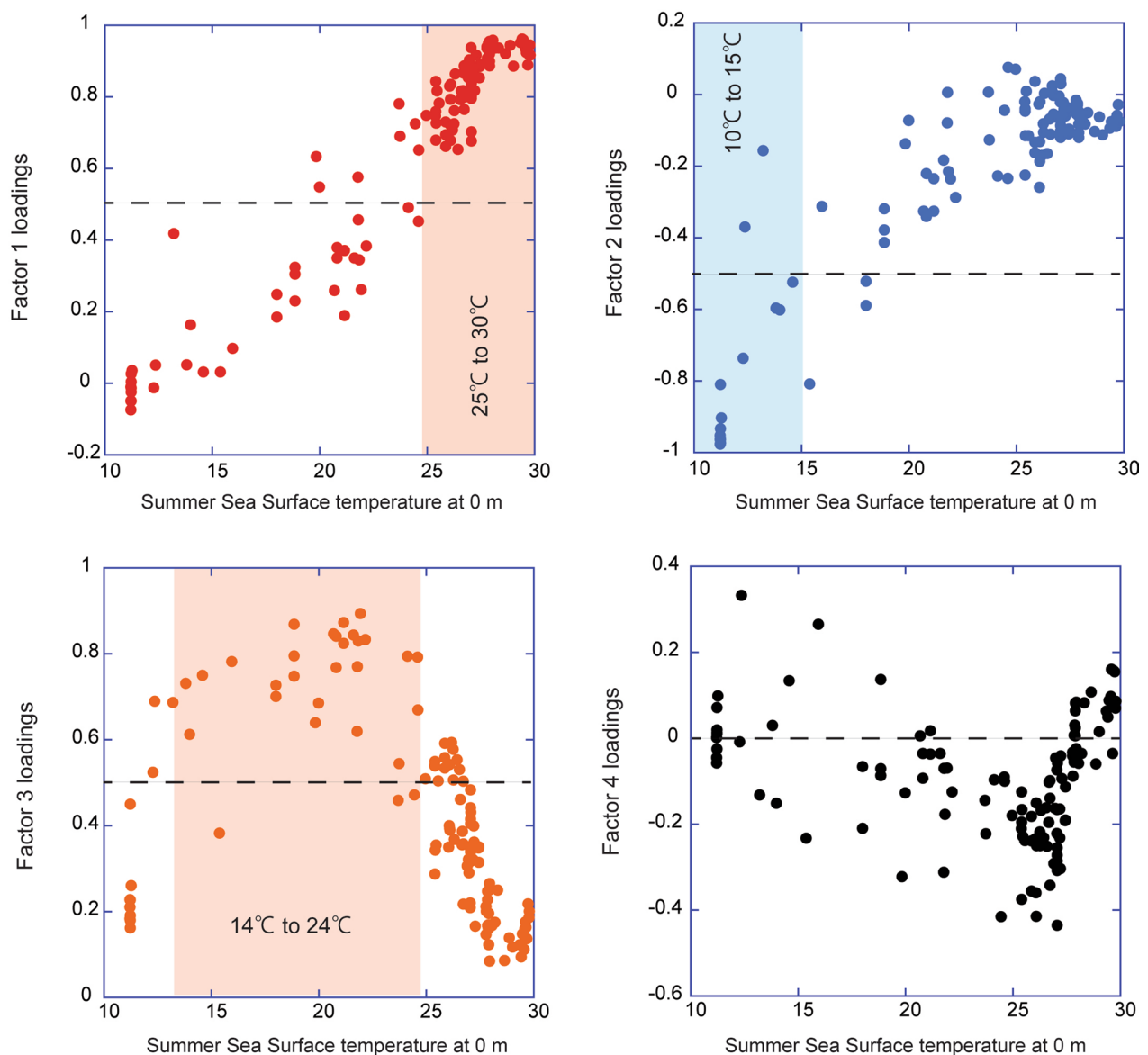


Fig. 6 Results from Factor Analysis conducted on core-top radiolarian assemblages used in this study. Plots show factor loadings of each radiolarian factor against summer SST. The temperature data are summer SSTs at 0 m of water depths from the World Ocean Atlas 2018 (Locarnini et al. 2018)

to *L. boreale*, *Antarctissa?* sp.1 is abundant in the Sea of Okhotsk, making up approximately 10% of the population. However, unlike *L. boreale*, *Antarctissa?* sp.1 is exclusive to the Sea of Okhotsk (Fig. 8). *Antarctissa?* sp.1 is associated with the Sea of Okhotsk Dicothermal Layer, which forms when sea ice melts during summer, creating a thin, warmer surface water layer (Kitani 1973; Yang and Honjo 1996; Abelman and Nimmergut 2005). This may explain why *Antarctissa?* sp.1 is endemic to the Sea of Okhotsk, suggesting that it may grow in response to the formation and disappearance of the Sea of Okhotsk Dicothermal Layer. Considering the results of earlier research,

the hypotheses presented by Matsuzaki and Itaki (2017), and the recent data described in this paper, it appears that *Antarctissa?* sp.1, *C. borealis*, and *L. boreale* all correspond to subarctic water masses related with the Sea of Okhotsk. Specifically, *Antarctissa?* sp.1 seems to be endemic to the Sea of Okhotsk, whereas *C. borealis* can also be founded in area associated with the OC.

2.3.3 Oyashio current-related subarctic to temperate assemblages

The Factor 3 exhibits strong loadings (greater than 0.5) between latitudes 35°N and 50°N, with the exception of

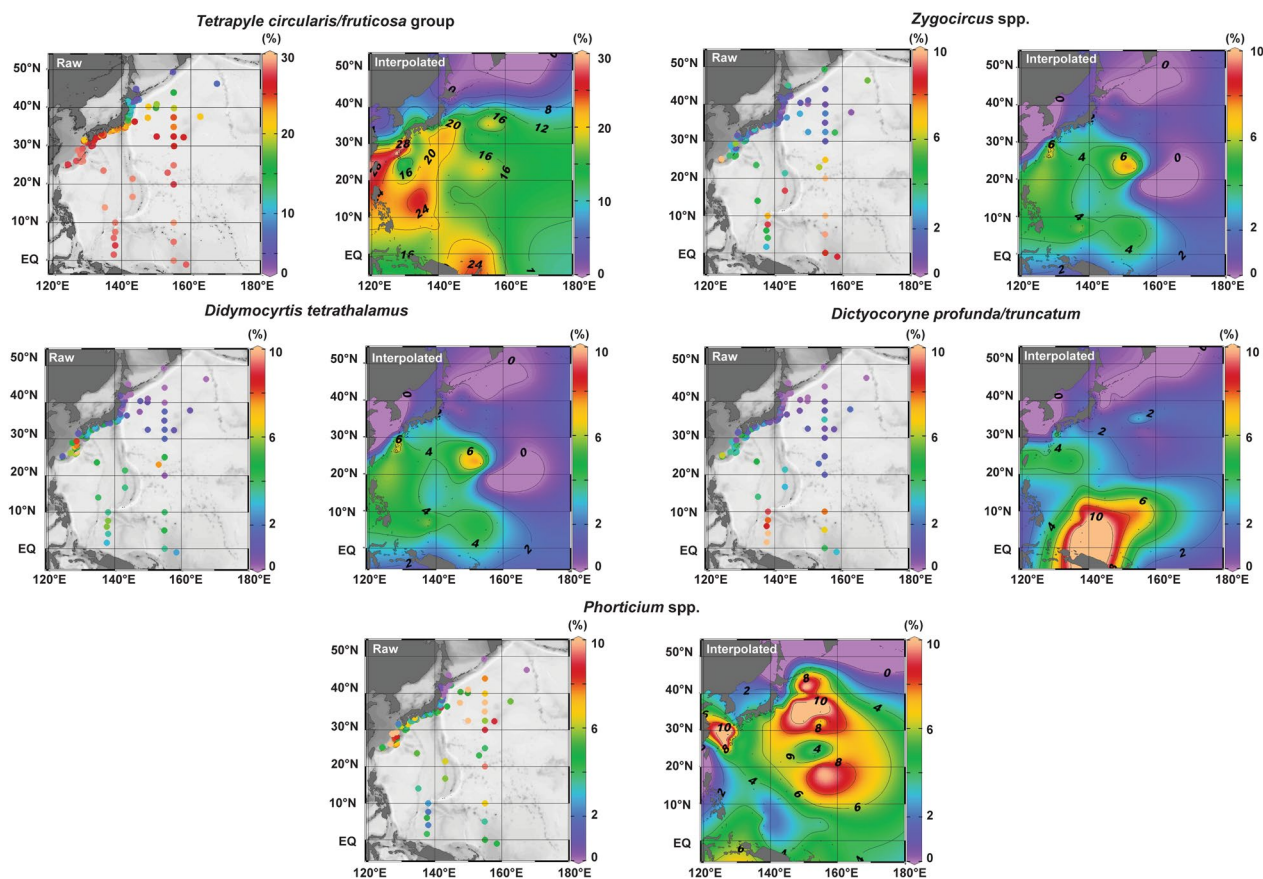


Fig. 7 A map showing the spatial distributions of key species having high scores in F1. The maps are sourced from ODV4 by Schlitzer et al. (2022). The left panels show raw loadings data, and the right panels show raw loadings data using DIVA (Data-Interpolating Variational Analysis) gridding interpolation (Schlitzer et al., 2022)

the Sea of Okhotsk (Table 3; Fig. 5). The peak loading occurs twice, (1) off the Sanriku Coast in northeastern Honshu, an area where the Kuroshio Current (KC) mixes with the Oyashio Current (OC), and (2) between latitudes 40°N and 50°N of the Northwest Pacific (Exclude Sea of Okhotsk) (Fig. 5). This factor shows high loadings for summer SSTs ranging from 25 °C to below 14 °C, suggesting an association with temperate to subarctic SSTs (Fig. 6). Key species for this factor, showing the highest factor scores, include *T. buestchlii* (3.3), *S. venustum* (2.2), *S. resurgens* (2.0), and *P. gracilipes* (1.9) (Table 4; Fig. 5).

T. buestchlii, *S. venustum*, *P. gracilipes*, and *S. resurgens* exhibit similar distribution patterns with high relative abundances (> 5%) in two key areas: (1) around latitude 40°N, off the Sanriku Coast in northeastern Japan and (2) in the open Northwest Pacific between latitudes 35°N to 50°N (Fig. 9). These species have been observed in sediment traps, surface sediments, and plankton samples from the subarctic Pacific, a region influenced by the Oyashio Current and its associated subarctic currents

(e.g., Nigrini 1970; Kling 1979; Nimmergut and Abelman 2002; Okazaki et al. 2004; Abelman and Nimmergut 2005; Kamikuri et al. 2008).

The springtime influx of *S. venustum*, *T. buestchlii*, and *S. resurgens* in the waters off northeastern Japan is thought to be linked to the inflows of the Oyashio Current (Itaki et al. 2008). Therefore, Species associated with Factor 3 are likely sensitive to cold and nutrient-rich, shallow waters related with the OC.

However, these species also reach peak abundances around latitude 40°N off the Sanriku Coast, an area influenced by the mixing of the Kuroshio Current (KC) and Oyashio Current (OC). The Sanriku Coast is characterized by the convergence of subtropical and subarctic waters transported by the KC and OC, respectively (Fig. 2). The frequent vertical mixing of these water masses, due to density differences, results in high nutrient enrichment (Talley et al. 1995; Itoh et al. 2015). Therefore, we suggest this enhanced productivity creates nutrient-rich conditions and relatively cold waters,

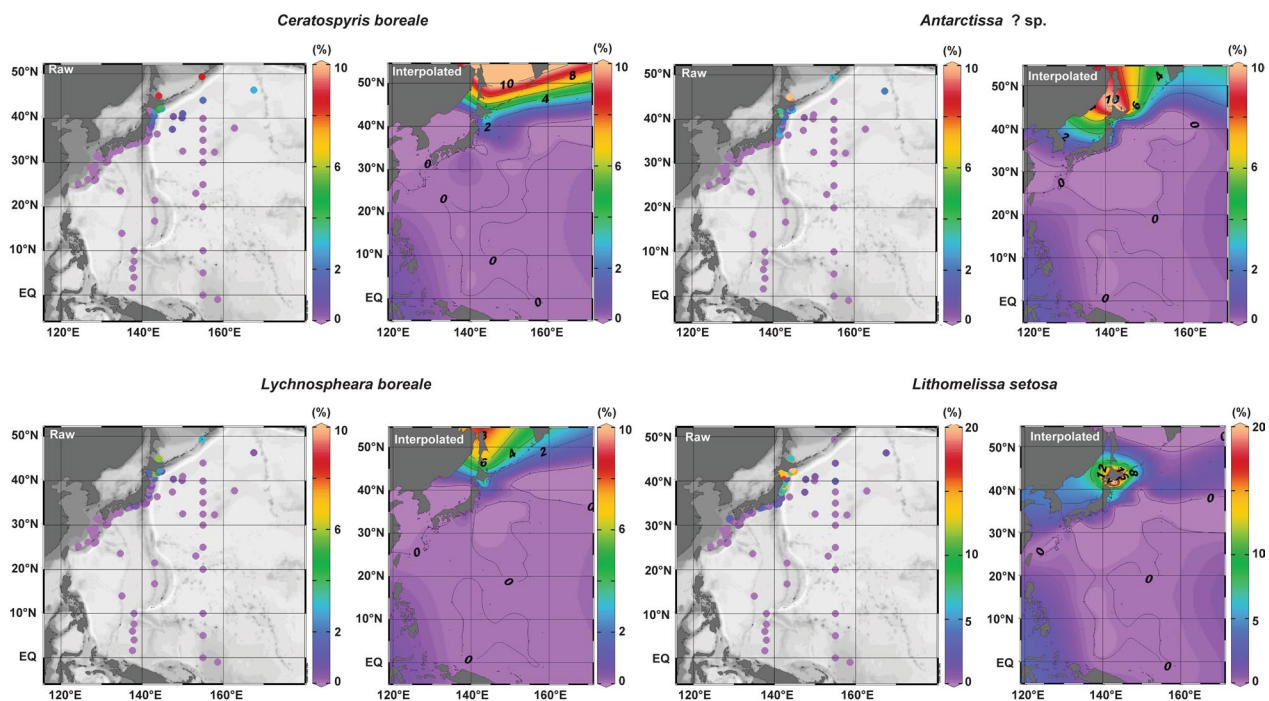


Fig. 8 A map showing the spatial distributions of key species having high scores in F2. The maps are sourced from ODV4 by Schlitzer et al. (2022). The left panels show raw loadings data, and the right panels show raw loadings data using DIVA (Data-Interpolating Variational Analysis) gridding interpolation (Schlitzer et al., 2022)

likely favoring the bloom of OC related species like *T. buetschlii*, *S. venustum*, *P. gracilipes* and *S. resurgens*.

2.3.4 Warm coastal assemblages

Factor 4 exhibits a distinct pattern linked to the coastal waters of southern Japan, as showed by moderately high factor loadings (0.4) in this region (Table 3, Fig. 5). This suggests a strong influence of coastal processes on this factor. Species with high factor scores for Factor 4 include *A. calvata* (− 2.3), *B. auritus* (− 2.0), and *L. erosa* (− 1.7) (Table 4), indicating their affinity for these coastal environments (Fig. 10).

Interestingly, these species do not exhibit a clear relationship with sea surface temperature (SST) variations in the Northwest Pacific (Fig. 6). This lack of a clear SST signal suggests that other factors, such as freshwater influence, might be more critical in controlling their distribution patterns. Indeed, Matsuzaki et al. (2016) identified *A. calvata* as a species dominant in the East China Sea Continental Shelf, an area known for freshwater influence. Given the potential influence of freshwater and the absence of a clear SST affinities, we excluded the species associated with Factor 4 from the SST reconstructions presented in the next chapter. This decision ensures that our SST estimates are not biased by factors unrelated to temperature variations.

2.4 Radiolarian-based SST estimates in the Northwest Pacific and their application to IODP site U1429 (East China Sea): suitability and limits

2.4.1 Application of updated methodology and re-estimation of radiolarian-based SSTs at IODP site U1429 (East China Sea)

The Factor Analysis indicated that Factor 1 (subtropical), Factor 2 (Sea of Okhotsk and surrounding subarctic Northwest Pacific), and Factor 3 (Oyashio Current-related temperate to subarctic assemblages) are likely related to SST ranges, while Factor 4 show poor affinities with SSTs. Species with high factor scores in Factors 1 to 3 demonstrate strong sensitivity to SST variations, making them suitable for SST reconstruction. Species with factor scores above 1 in these Factors are statistically significant drivers of the SST ranges associated with their respective Factors. Therefore, all species having factor scores higher than 1 in Factor 1 to 3 were selected for a WA-PLS, to reconstruct Summer SSTs at IODP Site U1429 located in the northern East China Sea. These species are *S. resurgens*, *S. venustum*, *T. buetschlii*, *L. boreale*, *Antartissa* (?) sp.1., *C. borealis*, *D. profunda/truncatum*, *D. muelleri*, *D. tetrathalamus*, *Phortitium* spp., *P. lappacea/spinosa*, *L. setosa*, *B. scutum*, *R. medianum*, *A. vinculata*, *Dictyocryphalus hispida*, *D.*

Table 4 Factor scores obtained for the selected taxa in the reference dataset from core-top samples

Species names	Factor 1	Factor 2	Factor 3	Factor 4
<i>Spongodiscus resurgens</i>	0.54	0.40	2.03	0.13
<i>Stylochlamyidium venustum</i>	−0.50	0.25	2.22	1.76
<i>Tholospira buetschlii</i>	−0.05	0.74	3.27	0.24
<i>Lychnosphaera boreale</i>	−0.11	−1.85	0.00	0.06
<i>Antarctissa?</i> sp.	−0.20	−2.62	0.27	0.58
<i>Ceratospyrus borealis</i>	−0.16	−1.80	0.41	0.84
<i>D. profunda/truncatum</i>	1.69	−0.08	−0.20	1.46
<i>Dictyocoryne muelleri</i>	1.10	−0.08	0.16	0.42
<i>Didymocyrtis tetrathalamus</i>	1.74	−0.12	−0.21	0.71
<i>Disolenia</i> spp.	0.64	−0.13	−0.28	0.32
<i>Heliodiscus asteriscus/echiniscus</i>	0.32	0.04	0.04	0.26
<i>Heliodiscus macrococcus</i>	0.20	0.01	−0.01	0.13
<i>Phortidium</i> spp.	1.79	0.29	0.62	−1.39
<i>Polysolenia lappacea/spinosa</i>	1.10	−0.18	−0.40	0.79
<i>Spongaster tetras tetras</i>	0.53	−0.01	−0.08	0.59
<i>Spongosphaera streptacantha</i>	0.01	−0.03	0.09	−0.28
<i>Lithomelissa setosa</i>	−0.45	−3.52	0.84	−1.30
<i>Botryocyrtis scutum</i>	1.01	−0.12	−0.26	0.89
<i>Rhizosphaera medianum</i>	−0.26	0.43	1.41	0.50
<i>Acanthodesmia vinculata</i>	1.06	−0.15	−0.36	0.19
<i>Dictyocryphalus hispida</i>	0.44	0.01	0.12	−1.18
<i>Dictyopodium praetextum</i>	1.01	−0.01	−0.22	−0.66
<i>Pseudodictyophimus gracilipes</i>	−0.28	−0.74	1.97	−0.43
<i>Tetrapyle fruticosa/circularis</i> group	3.06	0.19	0.93	−0.91
<i>Zygocircus</i> spp.	1.85	−1.54	−0.46	1.15
<i>Stylodictya tenuispina/stellata</i> group	0.23	0.17	1.04	0.53
<i>Collosphaera tuberosa</i>	0.23	−0.05	−0.11	0.21
<i>Lampromitra erosa</i>	0.21	0.01	0.05	−1.74
<i>Botryostrobus auritus</i>	0.50	0.02	0.33	−2.06
<i>Archipерidium calvata</i>	0.31	−0.11	0.27	−2.30

Bold indicate species having factor scores higher than 1 in one of the 4 dominant factors

praetextum, *P. gracilipes*, *T. fruticosa/circularis*, *Zygocircus* spp., and *S. tenuispina/stella* (i.e., Fig. 2).

The results of this analysis show that, for the reconstructed SSTs based on the compiled 122 samples covering the open Northwest Pacific and the ECS, all the samples except two are within the 95% confidence ellipse, and the residuals show that the most frequency estimating errors are between −1.5 and +1.0 (Table 5; Fig. 11). Therefore, we consider that the updated radiolarian dataset we compiled and the revisited methodology proposed in this review are highly suited to the accurate reconstruction of past summer SSTs in the Northwest Pacific.

As shown in Table 6 and Fig. 12, we obtained an average radiolarian-based summer SST of 23.76 °C ($N=121$, $SD=1.3^{\circ}\text{C}$) for site U1429 between 400 and 0 ka. The minimum value of 20.38 °C corresponds to 156 ka, which was during Marine Isotopic Stage 6 (MIS6).

Comparing this with previous estimates of Matsuzaki et al. (2019), the reconstructed SST minima is 2 °C higher (18°C) in this study. The maximum of 26.74 °C occurred at 6 ka during MIS1 (Table 6; Fig. 12). Summer SSTs above 25 °C were also recorded for 1.5 ka (MIS1), 120 ka (MIS5), 211 ka (MIS7), and 294 ka and 324 ka (MIS9) (Fig. 12). In addition, we assessed the similarity between fossil and modern assemblages by calculating the Squared Chord Distance (SCD) using Past4 (Hammer et al., 2001). SCD values can range from 0.0 to 2.0, with 0.0 indicating identical proportions of species between the samples being compared. In our analysis, most samples showed SCD values below 1 (Fig. 13), indicating good similarity between fossil and modern assemblages at Site U1429. Therefore, our method for reconstructing past summer SSTs is likely robust.

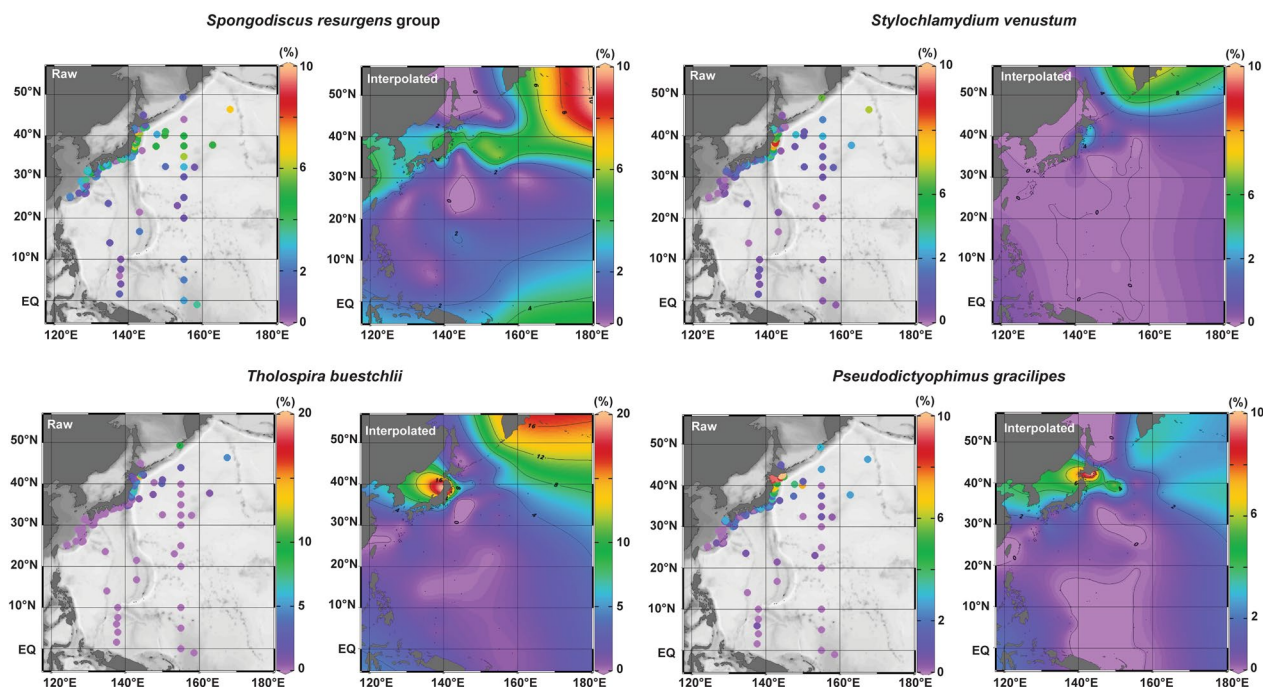


Fig. 9 A map showing the spatial distributions of key species having high scores in F3. The maps are sourced from ODV4 by Schlitzer et al. (2022). The left panels show raw loadings data, and the right panels show raw loadings data using DIVA (Data-Interpolating Variational Analysis) gridding interpolation (Schlitzer et al., 2022)

The modern summer SST at this site is 26.4 °C according to the World Ocean Atlas (Locarnini et al. 2018). Our reconstructed value closely matches this observation, considering an estimation error of only ± 1.4 °C. Modern SSTs at site U1429 are around 18.0 °C in winter (three months mean average value) and 22.2 °C in spring (three months mean average value) (Locarnini et al. 2018). Our results indicate that, even during glacial periods, summer SSTs were higher than 18 °C and were close to the modern spring value (22.2 °C) (Fig. 12). However, during Marine Isotope Stage 8 (MIS8), we estimate that summer SSTs were around 24 °C, which is only 2 °C less than the interglacial summer value (26.4 °C).

The MIS 8 is distinct from other glaciations that occurred within the past 400 kyr. At site U1429, benthic foraminiferal $\delta^{18}\text{O}$ values during MIS 8 shows comparable values to those of MIS 3 (Sagawa et al. 2018), suggesting that during MIS 8, either relatively warm deep water was present; the drop in sea level was less pronounced than during other glaciations, or a combination of both occurred. These conclusions are supported by the work of Siddall et al. (2007) and Elderfield et al. (2012), who suggested that there was a modest reduction in glacio-eustatic sea level during MIS 8 due to a lower global ice volume. The reasons behind these unique conditions during MIS 8 remain unclear; however, warming during this period was documented by Matsuzaki et al. (2014) for

the area off the Shimokita Peninsula on the northeastern coast of Japan. In contrast, a geological hiatus corresponding to MIS 8 has been found in sediment cores at IODP Expedition 315 sites C0001 and C0002, which are located off the coast of the Kii Peninsula, Japan (Matsuzaki et al. 2015a, b). In the Northwest Pacific, the Shatsky Rise ODP site 1209 shows a spike in primary productivity (Bordiga et al. 2013), while ODP site 1210 shows the establishment of modern water mass structures during MIS 8 (Chiyonobu et al. 2012). A warming trend has also been noted at North Pacific ODP sites 1143, 1146, and 806B (Lang and Wolff 2011). Therefore, the relatively high summer SSTs recorded during MIS 8 appear to be consistent with the global climatic context, suggesting that our revised method and re-estimated summer SSTs at Site U1419 are robust.

2.4.2 Comparison of radiolarian-based SSTs with geochemically reconstructed SSTs at IODP site U1429 (East China Sea)

To investigate how suitable radiolarian-based summer SSTs are for use with other proxies, we compared the radiolarian-based summer SSTs obtained in this study with the alkenone-based SSTs derived by Lee et al. (2021) and *Glogiberionoides ruber* Mg/Ca-based summer SSTs derived by Clemens et al. (2018) (Fig. 12).

Table 5 List of samples and residuals obtained during weighted averaging partial least squares analysis with two components

Samples	Summer sea surface temperature (°C)	Reconstructed	Residual
KH-22-7-OP-MC03 0–1 cm	29.79	27.98	–1.82
KH-22-7-OP-MC02 0–1 cm	29.77	28.09	–1.68
NH90-1-N54-NB17	29.74	28.25	–1.50
NH90-1-N56-NB18	29.68	29.02	–0.66
KH-22-7-OP-MC04 0–1 cm	29.64	28.56	–1.08
NH90-1-N57-NB19	29.61	28.47	–1.14
NH90-1-N60-NB21	29.58	28.64	–0.94
NH90-1-N62-NB22	29.56	29.06	–0.50
KH-22-7-OP-MC05 0–1 cm	29.47	28.42	–1.05
NH90-1-N68-NB25	29.42	28.59	–0.82
KH-22-7-OP-MC01 0–1 cm	29.36	28.28	–1.07
KH-22-7-OP-MC070–1 cm	29.01	27.88	–1.13
KR02-01 PC03-01 A	28.86	28.63	–0.22
KR13-02 PC03-01 W	28.64	28.53	–0.11
NH90-1-N79-NB30	28.32	27.56	–0.77
KH-21-3 PL04 0–2 cm	28.06	27.64	–0.42
KH-22-7-OP-MC08 0–1 cm	28.19	27.95	–0.24
KH-21-3 PL03 0–2 cm	27.90	28.47	0.57
KH-21-3 PL02 0–2 cm	27.90	27.77	–0.14
KH-21-3 PL01 0–2 cm	27.90	27.22	–0.68
GH08-198	27.94	27.24	–0.70
GH08-214	27.94	28.42	0.48
KH-22-3-PL08-0–2 cm	27.79	26.77	–1.02
GH08-197	27.91	27.57	–0.34
GH08-247	27.91	27.86	–0.04
GH08-248	27.88	28.12	0.24
GK14-612	27.84	28.09	0.25
GK14-642	27.84	27.62	–0.22
GK14-694	27.78	27.40	–0.38
GK14-591	27.85	27.36	–0.49
GK14-766	27.78	28.39	0.61
KS-22-4 MC04 0–1 cm	27.04	27.83	0.79
KS-22-4 MC07 0–1 cm	27.04	28.16	1.11
KH-22-3-PL01-0–2 cm	27.28	28.23	0.94
KH-22-7-OP-MC09 0–1 cm	26.67	26.36	–0.30
GH84-3-63	27.43	26.86	–0.57
GH84-3-169	27.44	27.56	0.11
GH84-3-30	27.44	27.37	–0.08
GH84-3-169	26.98	27.73	0.74
GH84-3-37	27.19	27.39	0.20
GH84-3-263	26.96	27.31	0.36
GH84-3-224	26.69	27.04	0.36
GH84-3-34	27.22	27.39	0.17
KH-21-3 MC06 0–1 cm	27.05	26.69	–0.36
KH-21-3 MC04 0–1 cm	27.05	26.93	–0.12
KH-21-3 MC03 0–1 cm	27.05	27.13	0.08
KH-21-3 MC01 0–1 cm	27.05	26.45	–0.60
KH-21-3 MC02 0–1 cm	27.05	25.19	–1.86
KH-21-3 MC05 0–1 cm	27.05	26.14	–0.91

Table 5 (continued)

Samples	Summer sea surface temperature (°C)	Reconstructed	Residual
GH83-2-320	27.19	27.39	0.20
GH83-2-264	27.21	26.83	-0.38
IODP-346-U1429 A-1H1-27-29	26.43	24.89	-1.54
KH-22-7-OP-MC10 0-1 cm	26.05	26.24	0.19
GH83-2-158	26.90	27.77	0.87
GH83-2-109	27.06	27.86	0.80
KH-22-7-OP-MC11 0-1 cm	26.09	25.95	-0.14
KH-22-7-OP-MC21 0-1 cm	26.30	26.36	0.06
GH82-1-24	27.05	25.12	-1.93
GH82-1-1	27.07	26.91	-0.16
GH82-1-49	26.73	27.74	1.01
GH82-1-51	26.73	26.77	0.05
GH82-1-69	26.73	26.36	-0.36
GH82-1-106	26.58	26.02	-0.57
GH82-2-110	26.07	25.68	-0.39
GH82-2-49	26.08	26.10	0.02
GH82-2-66	26.08	24.82	-1.26
GH97-258	25.55	25.51	-0.04
GH82-2-11	25.46	27.25	1.79
GH82-2-32	25.41	25.64	0.22
GH97-118	25.86	24.37	-1.48
GH97-126	25.86	24.75	-1.11
GH97-135	25.86	25.28	-0.58
GH97-163	25.41	25.41	0.01
GH97-175	25.41	24.23	-1.18
GH97-80	25.39	25.90	0.51
KR09-10 PC01-01 A	26.25	24.91	-1.35
GH80-2-409	26.54	25.58	-0.96
KH-22-7-OP-MC12 0-1 cm	24.96	25.32	0.36
GH80-2-407	26.27	25.90	-0.36
GH80-2-411	26.18	24.68	-1.50
KS-24-6-MC01-H-0-1 cm	24.44	25.87	1.43
MR01-ENG MC01-H01 W	25.40	26.15	0.75
GH81-2-271	24.12	21.08	-3.05
GH81-2-174	24.60	21.97	-2.64
YK21-07S-PL06 0-2 cm	23.74	22.78	-0.96
KH-22-7-OP-MC13 0-1 cm	23.70	25.39	1.69
MR98-05 MC06 (St.5)-H**W	24.62	24.83	0.21
GH81-2-113	22.16	20.61	-1.55
GH81-2-71	21.84	21.29	-0.55
GH81-2-44	21.61	21.77	0.16
GH81-2-30	21.94	20.13	-1.81
GH81-1-86	21.16	20.01	-1.15
GH81-1-69	21.16	20.34	-0.82
GH81-1-38	20.81	20.50	-0.31
GH81-1-30	20.81	19.96	-0.85
GH81-1-13	20.68	19.81	-0.87
KH-22-7-OP-MC14 0-1 cm	21.78	23.90	2.13
MR99-K04 MC02-H*1 W	21.79	20.91	-0.88

Table 5 (continued)

Samples	Summer sea surface temperature (°C)	Reconstructed	Residual
KR08-10 PC03-01 A	19.83	23.51	3.68
MR98-05 MC01 (St.1)-H**W	18.85	18.07	−0.79
GH82-3-13	18.84	21.13	2.29
GH82-3-16	18.84	20.80	1.96
KH-22-7-OP-MC18 0–1 cm	19.98	21.21	1.23
GH82-3-57	18.00	18.50	0.50
GH03-282	15.38	14.78	−0.60
GH82-3-111	18.00	20.30	2.30
GH03-200	14.58	15.30	0.71
KS-24-6-MC03-H-0–1 cm	13.99	17.06	3.07
GH03-102	13.81	15.35	1.54
GH03-109	12.28	14.53	2.24
KH-22-7-OP-MC15 0–1 cm	13.21	19.80	6.59
GH00-151	11.22	14.40	3.18
GH00-150	11.22	14.77	3.55
GH00-149	11.22	13.50	2.28
GH00-148	11.20	12.63	1.43
GH00-147	11.20	13.56	2.36
GH00-146	11.20	13.40	2.19
GH00-145	11.22	12.93	1.71
GH00-143	11.22	12.75	1.53
GH00-142	11.27	13.51	2.24
MR01-K03 MC03-H04 W	15.94	17.04	1.10
MR00-K03 MC04-H06 A	12.36	14.98	2.62

The average of the alkenone-based SSTs at Site U1429 was 21.8°C ($N = 1735$, $SD = 1.5^\circ\text{C}$) (Lee, et al. 2021; Fig. 12). The comparison between radiolarian and alkenone SSTs indicates that, even considering the error margins of both proxies, radiolarian-based summer SSTs tend to be around 2 to 3°C higher than the alkenone-based SSTs over the past 400 ka in general (Fig. 12). There is a pronounced spring bloom of alkenone-producing coccolithophores in April and May in the East China Sea around 20 m water depth (Lee et al. 2014, 2021; Kim et al. 2015; Ko et al. 2018). This finding would suggest that alkenone-based SST in the area around the East China Sea infers spring SSTs. However, comparison of SST proxy values for the core top from Site U1429 and Holocene sediment collected in the same area indicated that reconstructed alkenone-based SSTs at Site U1429 core-top and Holocene sediments are both around 23.4 °C (Lee et al. 2021). These temperatures are close to the annual mean SST (22.2 °C), although they are slightly higher than the annual mean value, which is probably related to the low production of alkenones during winter (December–February) (Lee et al. 2021). Therefore, it is likely that alkenone-based SSTs infer annual mean SST and are

sensitive to the conditions at water depths around 20 m of the water column. Therefore, the gap between radiolarian-based SSTs and those of alkenone can be simply explained by the seasonality. The radiolarian-based SSTs as demonstrated above infer summer SST at water depths between 0 and 50 m in the tropical North-west Pacific. Thus, considering that alkenone-based SST infer annual mean SST at ~20 m of water depths, although both proxies indicated water condition of same water depths, radiolarian is warmer because it is related to summer conditions. During MIS 5 and MIS 9, we observe that differences between alkenone-based annual mean SST and radiolarian-based summer SSTs are much smaller, although radiolarian-based SST remains higher (Fig. 12). Given that MIS 5 and MIS 9 are recognized as warmer interglacial periods (Lisiecki and Raymo 2005), we speculate that there may be lower seasonality during these stages. This would reduce seasonality and plausibly account for higher annual mean SSTs and thus explain the closer values observed in this study between alkenone-based annual mean SST and radiolarian-based summer SSTs.

Finally, we compared the radiolarian-based SSTs with the *G. ruber* Mg/Ca-based SSTs derived by Clemens et al.

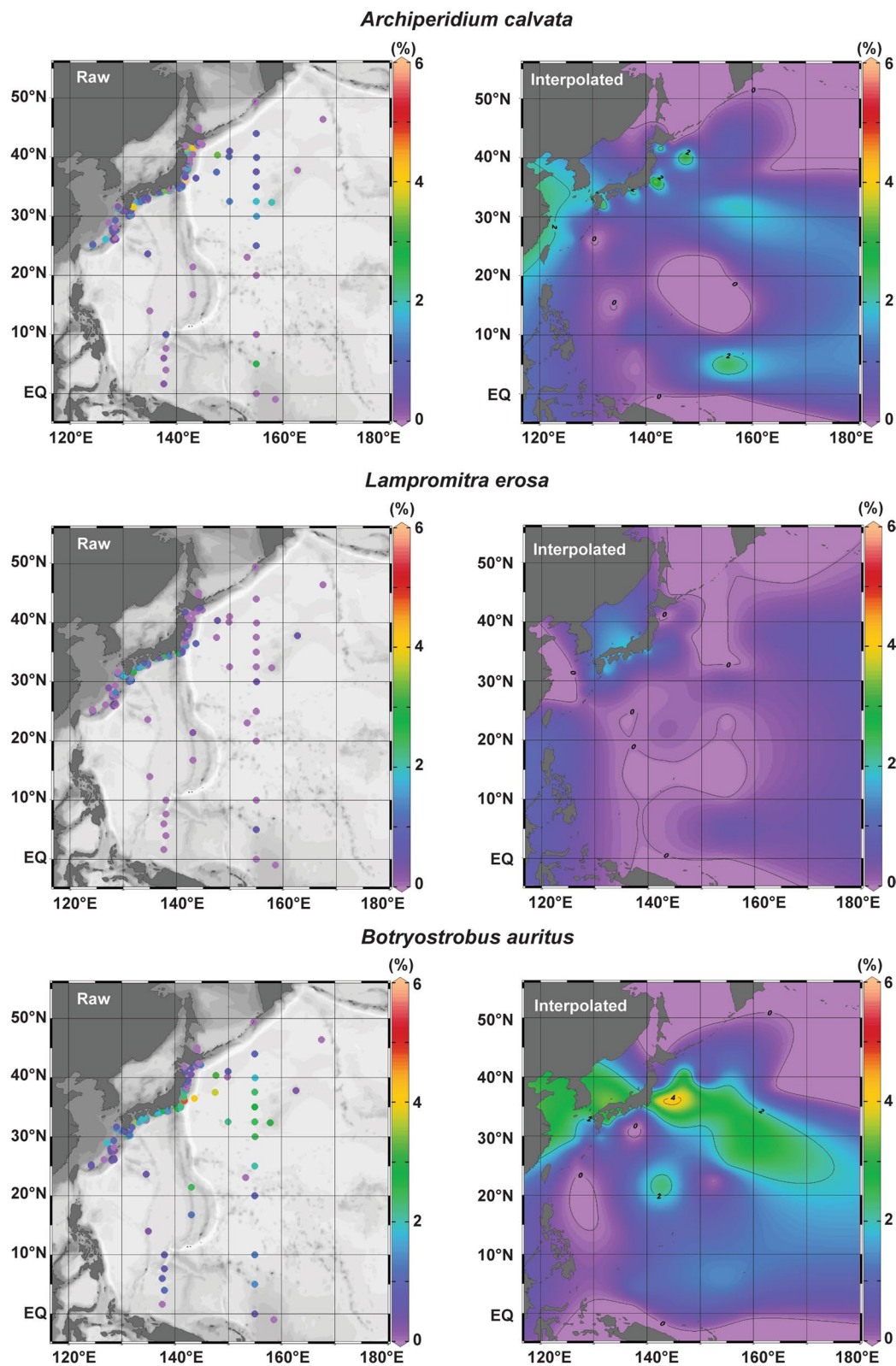


Fig. 10 A map showing the spatial distributions of key species having high scores in F4. The maps are sourced from ODV4 by Schlitzer et al. (2022). The left panels show raw loadings data, and the right panels show raw loadings data using DIVA (Data-Interpolating Variational Analysis) gridding interpolation (Schlitzer et al., 2022)

Table 6 Reconstructed radiolarian-based SST at IODP Site U1429 applying a weighted averaging partial least squares analysis with two components

Age (ka) of Sagawa et al. 2018	Revised radiolarian-based summer sea surface temperature (°C) (this study)
0.79	25.35
1.52	25.47
3.42	25.33
6.05	26.35
6.78	26.74
7.48	25.24
8.74	23.73
10.17	24.00
11.37	23.55
14.00	21.79
14.73	22.41
17.82	21.30
20.45	22.78
23.09	22.90
25.72	21.57
28.72	23.02
31.35	24.82
32.14	23.30
34.77	22.82
37.29	22.61
39.75	22.46
42.21	22.29
44.69	22.28
47.15	24.06
49.61	22.78
52.08	23.20
54.54	23.78
57.08	23.54
58.15	23.79
60.61	22.08
63.58	24.19
66.04	23.39
68.50	23.55
71.34	23.76
75.55	24.06
78.52	24.00
80.61	24.78
82.09	24.27
85.06	24.76
94.14	24.92
97.11	23.53
102.46	24.83
105.42	24.98
108.42	24.24
111.97	23.80
115.52	25.23
119.14	25.95

Table 6 (continued)

Age (ka) of Sagawa et al. 2018	Revised radiolarian-based summer sea surface temperature (°C) (this study)
120.12	25.83
122.69	24.88
126.24	23.33
129.79	24.28
130.50	22.70
134.05	21.87
137.60	21.92
141.15	21.14
144.70	23.16
147.96	20.96
149.12	20.78
152.67	22.81
156.32	20.38
159.17	22.69
162.01	22.54
164.77	22.33
167.67	23.64
170.52	21.65
172.81	24.68
175.66	24.15
178.53	22.97
180.52	23.09
182.17	23.99
185.04	23.76
187.88	22.98
190.73	25.42
195.68	24.18
199.20	25.51
202.72	24.40
206.24	24.95
211.36	26.03
214.88	25.82
218.40	24.79
221.90	23.67
228.85	23.27
232.37	25.26
235.91	25.37
238.33	24.67
239.99	23.97
251.82	23.06
254.33	23.86
258.31	22.80
260.82	23.41
263.33	22.42
265.84	22.35
270.85	22.56
275.81	24.49
288.24	25.32
290.37	24.22

Table 6 (continued)

Age (ka) of Sagawa et al. 2018	Revised radiolarian-based summer sea surface temperature (°C) (this study)
294.51	24.18
300.70	24.16
302.89	25.40
304.87	24.96
309.02	24.05
317.31	24.21
319.32	23.87
323.50	25.54
324.67	24.88
332.01	25.21
334.23	22.86
336.15	22.79
340.30	20.86
341.44	21.89
344.44	24.11
348.58	23.05
356.73	22.59
358.92	23.08
363.06	25.13
367.15	24.96
368.95	24.92
373.09	23.89
377.24	24.74
382.43	25.04
386.57	24.83

(2018). These *G. ruber* Mg/Ca-based SSTs have an average value of 23.85 °C ($N=1050$; $SD=1.5$ °C) (Fig. 12). *G. ruber* Mg/Ca-based SSTs fluctuate in a similar manner to the radiolarian-based summer SSTs. Unlike the alkenone-based SST estimates, the reconstructed *G. ruber* Mg/Ca-based SSTs and the radiolarian-based SSTs show similar values from MIS 1 to the onset of MIS 11 (Fig. 12). Indeed, when considering the error margins of radiolarian-based summer SSTs and *G. ruber* Mg/Ca-based SSTs, there is an excellent fit between both curves from 0 to 400 ka (Fig. 12). Traditionally thought to be a single species blooming in warm surface waters, a study by Wang (2000) identified two morphotypes of *G. ruber*, in the South China Sea, *G. ruber* s.s. (sensu stricto) and *G. ruber* s.l. (sensu lato) that have different stable isotopic compositions, which suggests they live at different ocean depths. Kuroyanagi and Kawahata (2004) reviewed Wang (2000) classification and focused on the differences between the two morphotypes, especially in terms of depth distribution in the seas around Japan, an understanding of these differences is critical to their use as proxies in marine environmental research. Kuroyanagi and Kawahata (2004) indicated that *G. ruber* s.s. peaks in abundance at depths of 0–20 m and is the predominant form in water samples collected close to the surface in areas influenced by the KC. In contrast, *G. ruber* s.l. has maxima in its abundance at both 0–20 m and the deeper range of 60–120 m in the same areas (Kuroyanagi and Kawahata 2004). This suggests the two morphotypes have different ecological preferences: *G. ruber* s.s. appears to rely on its symbiotic relationship with surface-dwelling algae,

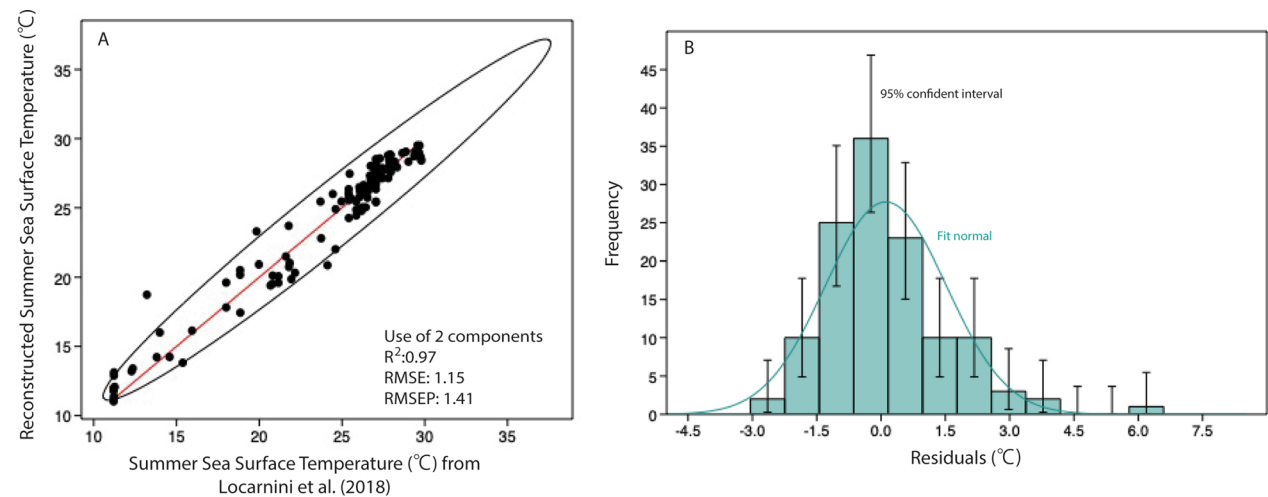


Fig. 11 Statistics of the weighted averaging partial least squares (WA-PLS) regression used in this study and the distribution of the obtained residuals

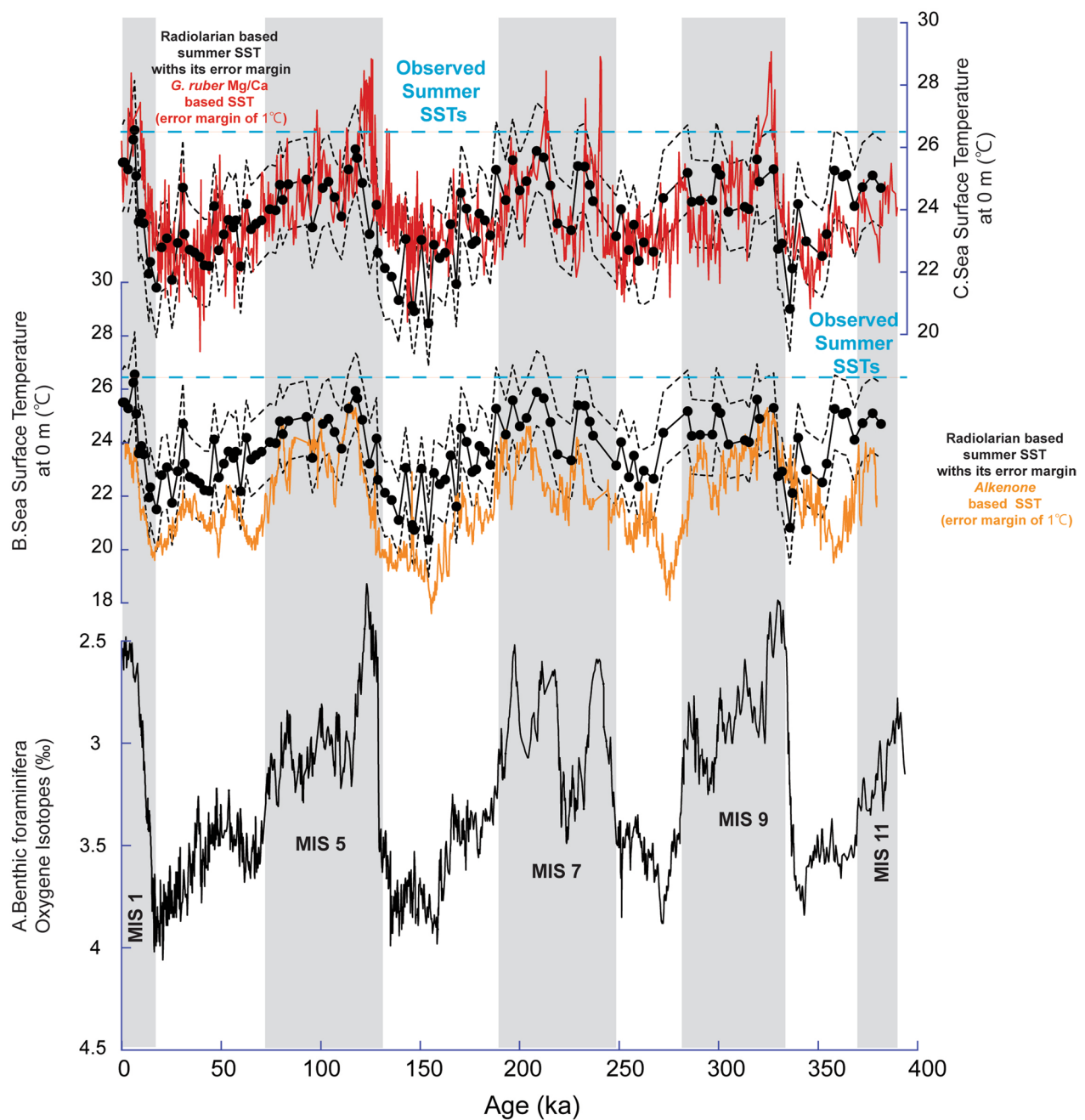


Fig. 12 **A** benthic foraminifera oxygen isotope (‰) stratigraphy at Site U1429 from Sagawa et al. (2018); **B** radiolarian-based reconstructed summer mean SST (°C) at Site U1429 (this study) vs alkenone-based SST (°C) of Lee et al., (2021); **C** radiolarian-based reconstructed summer mean SST (°C) at Site U1429 (this study) vs *G. ruber* Mg/Ca-based SST (°C) of Clemens et al. (2018). Dashed lines indicate the error margin of radiolarian-based summer SSTs (1.41 °C)

which need optimum temperature and light conditions, whereas *G. ruber* s.l. appears more adapted to deeper zones beyond the euphotic layer and may rely more on feeding than on symbiosis (Kuroyanagi and Kawahata 2004). In addition, data obtained from sediment traps located in the central Okinawa Trough suggest that the

seasonal fluxes of *G. ruber* are high between May and August (Xu et al. 2005). Consequently, Clemens et al. (2018) selected only *G. ruber* s.s. for Mg/Ca analysis, and thus the *G. ruber* Mg/Ca-based SSTs they obtained are related to summer SSTs at water depths between 0 and 20 m (Kubota et al. 2010; Clemens et al. 2018; Lee et al.

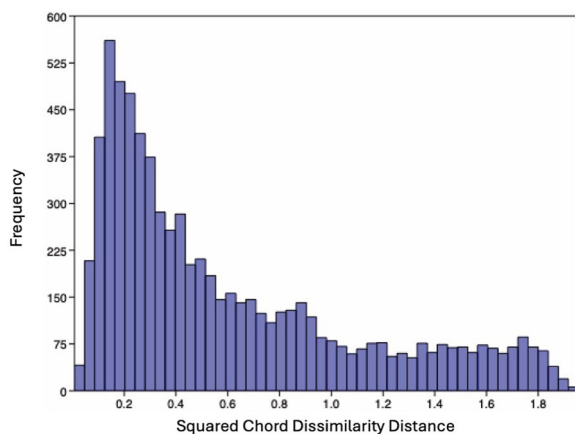


Fig. 13 Similarity between fossil and modern (downcore) assemblages calculating the Squared Chord Distance (SCD) using Past4 (Hammer et al., 2001)

2021). Therefore, the excellent fit between our radiolarian-based and *G. ruber* Mg/Ca-based summer SSTs is because both proxies, at least in the East China Sea, are related to summer conditions at water depths between 0 and 20 m. We conclude, therefore, that our method of estimating summer SSTs based on radiolarian assemblages is robust, and that the updated reference dataset for the Northwest Pacific and the simple statistical methods that were applied are appropriate.

However, some minor differences are observed during the coldest periods of MIS 2 (~ 20 ka), MIS 6 (~ 150 ka), and MIS 10 (~ 340 ka), where radiolarian-based summer SSTs show minor disparities with those from *G. ruber* Mg/Ca (Fig. 12). At site U1429, during these coldest periods, the subarctic coastal radiolarian species *L. setosa* dominated (Matsuzaki et al. 2019). *L. setosa* inhabits water depths between 50 and 100 m based on the supplementary data of Matsuzaki et al. (2016) in the East China Sea, and in northern Japan, *L. setosa* blooms in December when lithogenic fluxes are highest. Generally, lithogenic fluxes peak either during the summer monsoon, when there is high precipitation and strong riverine runoff, or during the winter monsoon, when dry continental winds transport suspended lithogenic matter such as dust (e.g., Lou et al. 2016). However, northern Japan lacks any major river systems that could deliver such high amounts of lithogenic fluxes, even during the summer monsoon. Thus, summer monsoon strong riverine runoff hypothesis is unlikely. Thus, we suggest that it is most likely that *L. setosa* to be associated with water depths around 50–100 m, and its productivity is enhanced during winter monsoons and related dust/nutrient deposition and *L. setosa* may be a valuable tool for monitoring winter monsoon variability in East Asian seas. Therefore,

because *L. setosa* dominates during MIS 2 (~ 20 ka), MIS 6 (~ 150 ka), and MIS 10 (~ 340 ka), it is likely that radiolarian-based SSTs during these periods reflect more winter temperatures at water depths between 50 and 100 m rather than surface summer SSTs, which may explain the disparities with *G. ruber* Mg/Ca. It is thus crucial to carefully consider assemblage composition and understand key species ecology when reconstructing SSTs based on fossil assemblages, as changes in inferred seasonality or water depths can occur when reconstructing SSTs over long time intervals. In particular, because the ECS is a shallow sea, the 100 m sea level fall which occurred during each glacial period may favor such significant changes in inferred SST seasonality, as documented by Kubota et al. (2019) and Matsuzaki et al. (2019).

3 Conclusions

In this study, we analyzed changes in radiolarian assemblages using 33 new samples collected in the Northwest Pacific and its marginal seas during Japanese scientific surveys since 2020 aboard the Hakuho-Mar, Shinsei-Mar, and Yokosuka. We combined these new data with the already published datasets of Matsuzaki and Itaki (2017), which covers the ECS and the Northwest Pacific Japanese coast, and of Motoyama et al. (2016), which covers the Sea of Japan. We reviewed and re-investigated the geographical distributions of different radiolarian species and their affinity with different water masses and re-estimated sea surface temperatures based on the results of the radiolarian species distributions. These estimates were then compared with reconstructed sea surface temperatures (SSTs) based on biogeochemical methods at IODP site U1429.

Through Detrended Correspondence Analysis, radiolarian assemblages of the Japan Sea showed different distribution patterns from those of the Northwest Pacific, likely reflecting the influence of regional oceanographic processes. We therefore excluded the Sea of Japan data from the subsequent reconstruction of SSTs. We then focused on radiolarian assemblages from the open Northwest Pacific and the East China Sea, where radiolarian assemblages respond to environmental changes linearly (almost). In these samples, we applied Factor Analysis and identified four main radiolarian assemblages in the Northwest Pacific: a subtropical assemblage, a Sea of Okhotsk-related subarctic assemblage, an Oyashio Current to transitional zone-related assemblage and coastal water assemblage.

In the subtropical assemblage, species such as *T. circularis/fruticosa* and *D. tetrathalamus* were found to be associated with the Kuroshio Current (KC). *T. circularis/fruticosa* exhibited specific distribution patterns

associated with warm nutrient-rich waters, whereas *D. tetrathalamus* was found to be associated with warm oligotrophic waters. Based on our results, *Phortidium* spp. can be indicators of transitional and warm water masses.

In the Sea of Okhotsk-related subarctic assemblage, species such as *L. setosa*, *C. borealis*, *L. boreale*, and *Ant-artictissa?* sp.1 were found to be abundant. Our results showed that the Sea of Okhotsk is dominated by *Ant-artictissa?* sp.1, which is endemic there due to the unique environmental conditions, including the dicothermal layer created by summer sea ice melt. The Oyashio Current to transitional zone-related assemblage included species such as *S. venustum*, *S. resurgens* and *P. gracilipes* and corresponded mainly to the Oyashio Current waters. Factor Analysis performed between the abundances of selected radiolarian species and summer SSTs revealed that factors 1 to 3 (subtropical to Oyashio Current) can be associated with different SST ranges. For example, assemblage including the species *T. circularis/fruticosa* and *D. tetrathalamus* were found to be associated with summer SSTs higher than 24 °C, whereas assemblage including *L. setosa* and *C. borealis* were found to be associated with summer SSTs lower than 14 °C.

Lastly, based on this analysis, a weighted averaging partial least squares analysis was conducted to reconstruct past summer SSTs in the ECS at IODP Site U1429, which resulted in accurate estimates with small prediction errors (1.4°C) and a R^2 of 0.97. We applied this revisited method at site U1429 (East China Sea). Radiolarian-based summer SST reconstructions at site U1429 (East China Sea) over 400–0 ka averaged 23.76 °C, ranging from 20.38 °C (MIS6) to 26.74 °C (MIS1). The reconstructions closely match modern observations (26.4 °C) and reveal unusually warm conditions during MIS8 (24 °C). This warmth during MIS8 is corroborated by evidence of reduced glaciation and warmer conditions across multiple Pacific sites. Based on these results, we consider that the new methodology described in this paper is suited to the accurate reconstruction of past summer SSTs in the Northwest Pacific and could provide valuable insights into radiolarian distribution patterns in the region.

We showed that the difference between the SSTs derived using the alkenone-based and radiolarian-based methods at IODP site U1429 could be explained by considering that radiolarian-based SSTs represent summer temperatures whereas alkenone-based SSTs correspond to annual mean temperatures. A close match was found between the radiolarian-based SSTs inferring summer SSTs and the *G. ruber* Mg/Ca-based summer SSTs, which again indicated the robustness of our method. However, minor disparities were observed between radiolarian-based and *G. ruber* Mg/

Ca summer SST reconstructions during the coldest periods of MIS 2, 6, and 10 (Glacial Periods). During these intervals, the subarctic coastal species *L. setosa* was dominant. *L. setosa* is abundant during winter monsoons and is associated with nutrient-rich shallow waters. Thus *L. setosa* dominance during the cold periods of MIS 2, 6, and 10 can indicate that the radiolarian-based SSTs likely reflect more winter temperatures at 50–100 m depth rather than surface summer SSTs during glacial periods cooling maxima in the East China Sea. This emphasizes the importance of considering species ecology when reconstructing palaeoceanographic conditions. Despite the differences between the datasets that were observed, we conclude that the overall outcomes of our applied method remain credible, allowing for an estimated error margin of 1.4 °C.

Abbreviations

CLIMAP	Climate Long-range Investigation, Mapping, and Prediction
NPEC	North Pacific Equatorial Current
KC	Kuroshio Current (KC)
ECS	East China Sea
OC	Oyashio Current
KCE	Kuroshio Current Extension
NPIW	North Pacific Intermediate Water
OSIW	Okhotsk Sea Intermediate Water
JSPW	Japan Sea Proper Water
CDW	Changjiang Diluted Water
PCW	Pacific Central Water
MIS	Marine Isotope Stage

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40645-025-00706-6>.

Supplementary material 1

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Author contributions

MK and IT co-proposed the topic and conceived and designed the study. MK conducted the reviews and radiolarian assemblage analysis, performed statistical analysis, and wrote the draft of the manuscript. LK, KY, and MI provided access to published datasets and assisted MK in interpreting them. ST, HK, MM, and OH were members of the GEOTRACES KH-22–7 expeditions and conducted essential core-top coring work for this study. ST collaborated with the corresponding author in manuscript construction. All authors have reviewed and approved the final manuscript.

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Availability of data and materials

The raw data are attached here as a supplementary data.

Declarations

Competing interests

The authors declare that they have no competing interest.

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